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The Evolution of Quantitative Equity Investing: Half a Century of Alpha Research and Institutional Adaptation

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ABSTRACT

This survey interprets the evolution of quantitative equity investing as a history of changing constraints rather than a succession of increasingly complicated forecasting models. Portfolio theory made diversification calculable; characteristic-based investing converted value, momentum, profitability, and low-risk effects into scalable processes; alternative data shifted competition toward information engineering; machine learning expanded the admissible class of return functions; and large language models and agentic systems are beginning to reorganize research itself. Yet each advance has exposed a new implementation boundary. A predictor must survive point-in-time reconstruction, model selection, portfolio translation, turnover, transaction costs, shorting frictions, capacity, latency, regime change, and institutional governance before it becomes alpha. The survey develops alpha decay as the mechanism linking these generations and gives particular attention to the limited transfer of machine-learning research into production. It argues that artificial intelligence will accelerate both discovery and commoditization. Sustainable advantage is therefore unlikely to reside in a generic model class. It is more likely to arise from proprietary information transformation, economic structure, cost-aware portfolio construction, execution, organizational memory, and tightly controlled human-machine systems.

JEL Classification: G11, G12, G14, G17

“The history of quantitative investing is not a story of models replacing one another, but of markets repeatedly revealing which assumptions the previous generation could no longer afford to ignore.”

1 | Introduction: Quantitative Investing as an Expanding Definition of the Problem

The conventional history of quantitative investing is a history of models. Markowitz is followed by the capital asset pricing model,

the capital asset pricing model by multifactor regressions, multifactor regressions by machine learning, and machine learning by foundation models. Such a chronology is not wrong, but it mistakes the visible mathematical instrument for the economic problem that led to its adoption.

Quantitative equity investing has progressed by discovering that the previous generation had defined the investment problem too narrowly. Mean-variance analysis formalized the trade-off between expected return and covariance, but it treated the relevant inputs as if they could be estimated with sufficient precision. The capital asset pricing model imposed equilibrium discipline on expected returns, yet empirical return differences

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associated with firm size, valuation, past performance, profitability, and investment were not absorbed by market beta. Factor investing systematized these patterns but often evaluated them in portfolios that abstracted from market impact, short-sale costs, capacity, and crowded liquidation. Alternative data broadened the information set but introduced new problems with historical coverage, licensing, entity matching, and revisions. Machine learning widened the function class but multiplied the routes through which leakage and model selection could enter. Large language models make unstructured information easier to process, while simultaneously making historical contamination, unstable inference, and persuasive but unsupported narratives harder to detect.

The history is therefore better expressed as a change in the object being optimized. Early empirical work asked whether expected returns could be represented as

$$E_t [r_{i,t+1}] = g(\mathbf{z}_{i,t}, \mathbf{m}_t),$$

where $\mathbf{z}_{i,t}$ denotes security characteristics and $\mathbf{m}_t$ denotes market conditions.

1.1 | Portfolio Expected Return and Alpha

Once security-level expected returns have been estimated, they can be aggregated into a portfolio forecast. Alpha is commonly measured relative to a benchmark factor model. In a simple representation, the intercept captures the component of return not explained by the benchmark factors.

$$E_t [r_{i,t+1} - r_{f,t}] = \alpha_{gross,i} + \sum_j \beta_j \cdot F_{j,t} + \varepsilon_{i,t}$$

1.2 | Interpretation and Scope

Here, r_f is the risk-free rate, β denotes the loading on a benchmark factor, and F denotes the corresponding factor return. The intercept, therefore, provides a benchmark-relative measure of gross alpha rather than a complete institutional definition of alpha. This representation is compatible with standard asset-pricing benchmarks, including the CAPM, arbitrage pricing theory, and Fama-French factor models (Fama and French 1993, 2015; Ross 1976; Sharpe 1964). The distinction matters because implementation costs and constraints subsequently determine how much gross alpha survives as net alpha.

Contemporary institutional investing must ask whether a forecast derived from that conditional expectation can be converted into a scalable position after all relevant frictions:

$$\alpha_{net}(A, t) = \alpha_{gross}(t) - C_{spread} - C_{impact}(A, t) - C_{borrow} - C_{delay} - C_{model} - C_{organisation}.$$

The final two terms deserve emphasis. C_{model} includes estimation error, instability, distribution shift, and specification uncertainty. $C_{organisation}$ captures losses caused by poor data lineage, slow approval processes, fragile infrastructure, unclear ownership, and the inability to diagnose or halt a system. These costs are rarely observable in public return databases, but they help explain

why an academically persuasive result may remain commercially unusable.

Figure 1 sets out the chronology adopted in this survey. The arrows should not be read as a technological replacement. Value did not cease to exist when momentum was documented, nor did profitability disappear when neural networks were introduced. Later generations absorb earlier ones and move the margin of competition elsewhere. A present-day machine-learning equity model may still derive most of its economic content from value, momentum, liquidity, volatility, and profitability. What has changed is the manner in which these components are conditioned, combined, traded, and monitored.

The movement from portfolio theory to agentic AI can consequently be understood as an accumulation of institutional layers. Portfolio theory supplied a language for covariance and diversification. Empirical asset pricing supplied characteristics and factor portfolios. Electronic markets and statistical arbitrage made execution and capacity explicit. Alternative data turned data provenance into an investment competence. Machine learning made validation architecture part of the investment model. Generative AI now makes the research organization itself an object of design.

This interpretation differs from a celebratory narrative in which each new method is presumed to dominate the previous one. The empirical record repeatedly warns against that presumption. DeMiguel et al. (2009) show how difficult it is for estimated optimal portfolios to beat the 1/N rule out of sample. Harvey et al. (2016) demonstrate that the accumulation of tested factors raises the evidentiary threshold for new discoveries. McLean and Pontiff (2015) find that anomaly returns decline out-of-sample and after publication. Gu et al. (2020) establish that nonlinear methods can improve return prediction, but the strongest gains do not imply that arbitrary depth or flexibility is useful. Jensen et al. (2026) show that the frictionless frontier generated by machine learning can be radically different from the frontier available to an investor paying realistic trading costs. These results do not reject quantitative innovation. They redefine the burden of proof.

The most contentious implication concerns the meaning of “alpha”. In academic usage, alpha is frequently an intercept relative to a benchmark model. In institutional usage, the term is closer to a scarce productive asset: a forecasting or trading process that improves the existing portfolio subject to risk, cost, capital, operational, and governance constraints. The two concepts overlap, but neither contains the other. A statistically significant intercept may be too expensive or too small to matter. A fund may possess a valuable execution or data-cleaning advantage that appears only as a modest increment to forecast accuracy. The survey, therefore, treats predictive fit as one input into alpha production, not as its definition.

Figure 2 maps the intellectual lineage behind this view. Modern financial machine learning is often presented as a departure from asset pricing, but it inherits two older research traditions. One concerns empirical return predictability, running from Fama–MacBeth regressions through characteristic portfolios and factor replication. The other concerns estimation and decision under

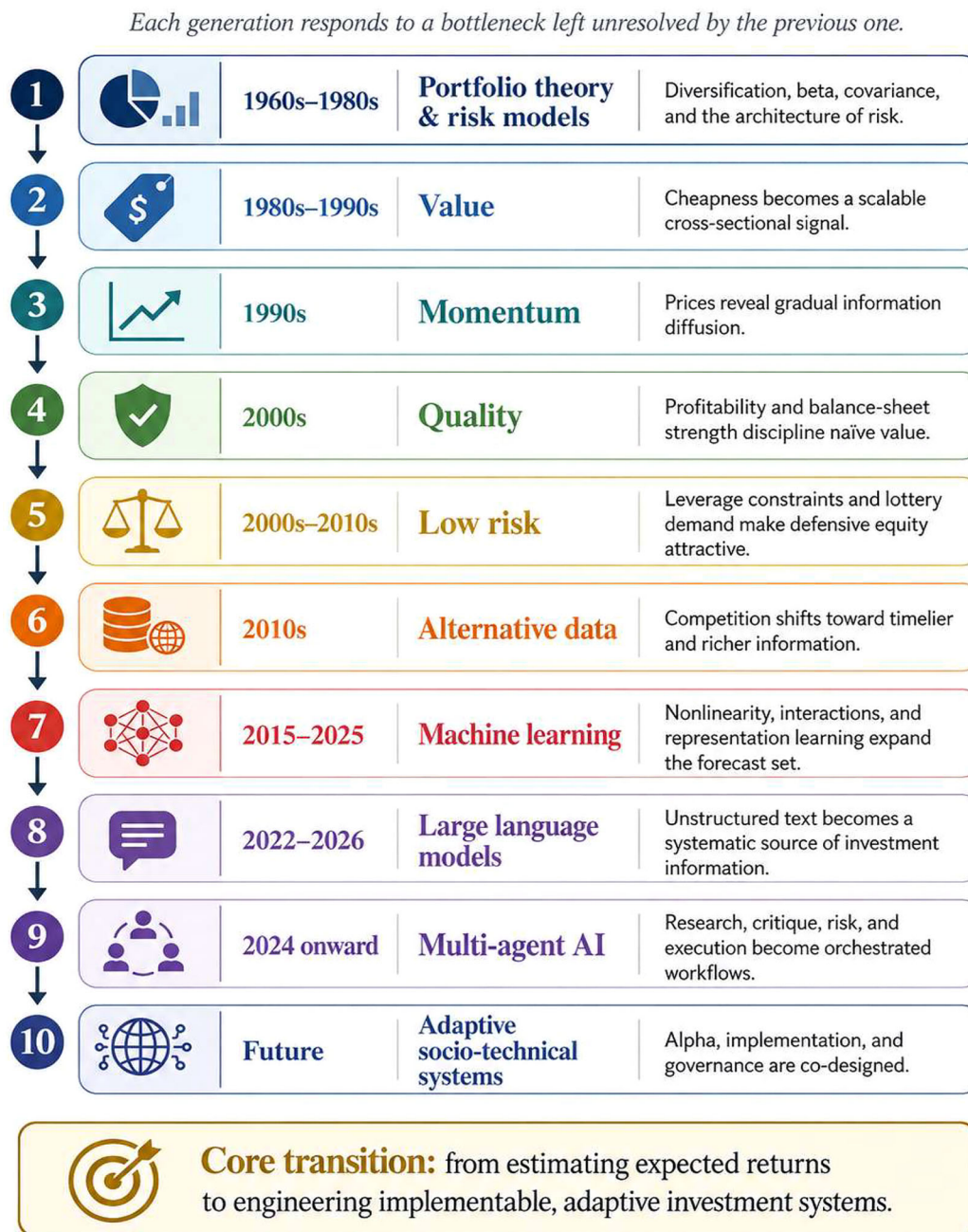


FIGURE 1 | Half a century of quantitative equity investing. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12028)]

uncertainty, running from Markowitz through shrinkage, naïve diversification, and dynamic trading with costs. The most promising implementation-aware machine-learning work sits at their intersection.

1.3 | Purpose of the Survey

A half-century survey necessarily condenses a wide range of heterogeneous strategies into broad generations. Table 1 makes this simplification explicit. Its purpose is not to impose sharp historical breaks, but to highlight the unresolved bottlenecks that shaped the direction of research and practice in each period.

1.4 | Continuity Across Investment Styles

The boundaries between these broadly accepted investment styles should be understood as general rather than fixed. In practice, the search for an explanatory edge in the cross-section of stock returns has driven a continuous and recursive process of innovation.

The remainder of the survey develops this argument in seven stages. The factor era is reconsidered as an institutional technology rather than a list of anomalies. Portfolio construction is then treated as the point at which academic and commercial notions of alpha diverge. Alpha decay provides the connecting mechanism. Alternative data is analyzed as an industrial information chain.

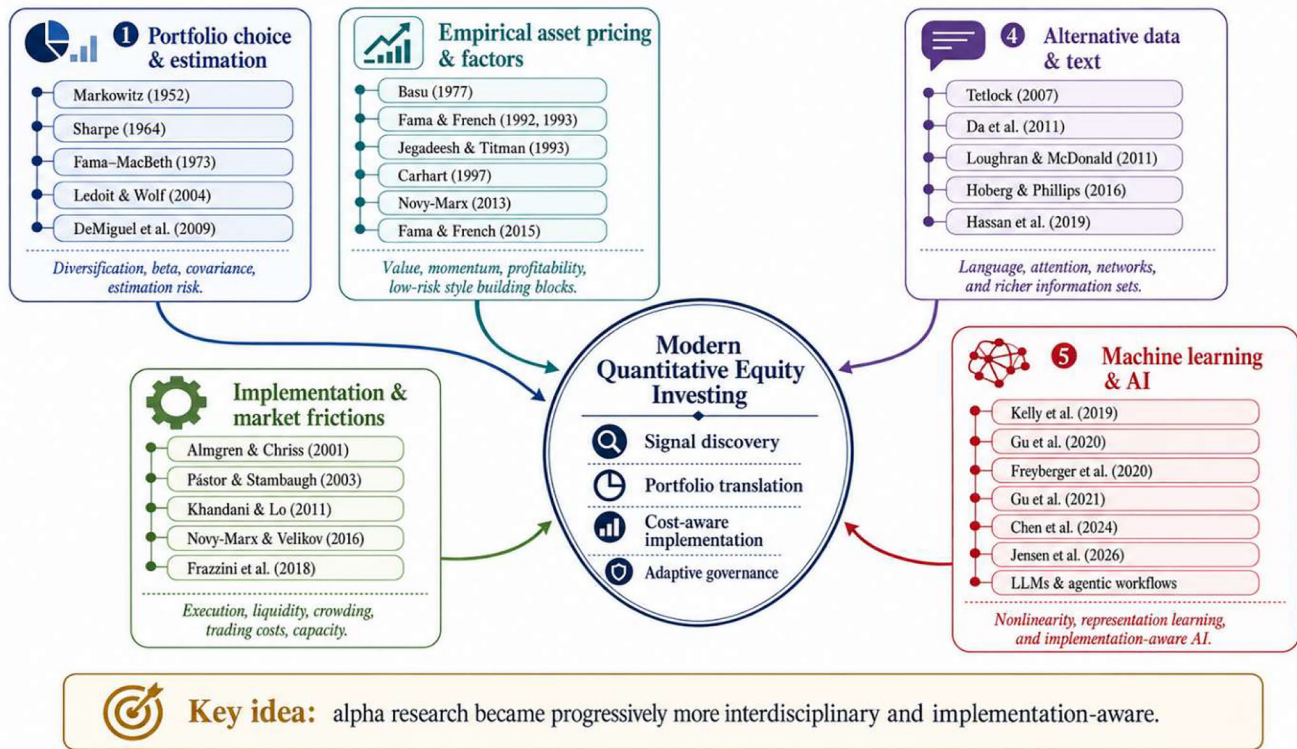


FIGURE 2 | Intellectual streams in quantitative equity research. [Colour figure can be viewed at wileyonlinelibrary.com]

Machine learning is evaluated through the implementation failures that separate a backtest from production. LLMs and multi-agent systems are assessed as research infrastructure rather than autonomous price oracles. The conclusion considers whether alpha can remain sustainable when discovery and imitation are both accelerated by artificial intelligence.

2 | The Factor Era: From Empirical Regularities to Institutional Products

2.1 | Portfolio Theory, Equilibrium, and the Empirical Turn

Markowitz (1952) changed investment analysis by making the portfolio, rather than the individual security, the unit of choice. The familiar optimization problem,

$$\min_w w' \sum w = \pi r^2 \quad \text{subject to } w' \mu = \mu^*,$$

is deceptively compact. Its institutional significance lay in the separation of expected return from covariance and in the recognition that an unattractive asset in isolation could improve a portfolio through diversification. Sharpe (1964) subsequently compressed systematic covariance into exposure to the market portfolio, while Fama and MacBeth (1973) provided an empirical method for estimating the cross-sectional relation between exposures and returns.

Recent work also questions the assumption that only systematic risk is priced. Dello Preite et al. (2025), for example, allow

expected returns to reflect compensation for unsystematic risk. Such extensions illustrate that even the expected-return representation remains an evolving object rather than a settled taxonomy.

The empirical difficulties emerged quickly. Expected returns are estimated with far less precision than variances, and covariance matrices become unstable as the number of assets grows. Small errors in $\hat{\mu}$ can create large and economically implausible positions. Ledoit and Wolf (2004) show why shrinkage of large covariance matrices improves conditioning, while Kan and Zhou (2007) demonstrate that parameter uncertainty materially changes the theoretically optimal portfolio. DeMiguel et al. (2009) supply the memorable practical benchmark: across a wide range of datasets and optimization rules, estimated portfolios did not consistently outperform equal weighting out of sample.

The $1/N$ result is occasionally misread as a defence of intellectual simplicity for its own sake. Its stronger implication is that every estimated degree of freedom must earn back its estimation cost. That principle later becomes central to machine learning. A flexible model enlarges the attainable class of forecasts, but it also enlarges the class of errors from which the researcher may select.

2.2 | Value: Systematic Contrarianism and Its Ambiguities

Value became quantitative when valuation ceased to be a narrative judgement and became a cross-sectional rule. Basu (1977) documented the association between earnings–price ratios and

TABLE 1 | Generations of quantitative equity investing.

Generation	Binding problem	Principal innovation	Typical fund implementation	Characteristic failure	Direction of the next upgrade
Portfolio theory and risk models	Diversifying risk and estimating covariance	Mean–variance analysis, beta, structured covariance models	Strategic allocation, benchmark-relative risk, optimization	Extreme sensitivity to expected-return and covariance estimates	Parsimony, shrinkage, empirical characteristics
Value and size	Market beta does not organize the cross-section	Characteristic sorts and multifactor models	Long cheap or neglected firms, short expensive firms	Value traps, accounting latency, sector and distress exposure	Momentum, profitability and composite signals
Momentum	Information is incorporated gradually	Cross-sectional ranking on past returns	Winner–loser books, residual momentum, volatility scaling	Turnover, crowded trends, severe rebound crashes	Conditional exposure and diversified signal ensembles
Quality and low risk	Naïve value purchases weak firms; leverage constraints distort pricing	Profitability, investment, defensive and low-beta factors	Quality–value combinations, defensive equity, constrained low-risk portfolios	Expensive defensiveness, duration exposure, speculative-market underperformance	Conditional factor construction and richer data
Alternative data	Conventional statements are delayed and coarse	Transactions, text, networks, search, imagery and digital exhaust	Point-in-time data pipelines linked to firms and revenues	Coverage drift, legal risk, revisions and rapid commoditization	Proprietary transformations and flexible representations
Machine learning	Linear additivity misses interactions and nonlinearities	Regularization, trees, neural networks, latent representations	Nonlinear rankings, ensembles, conditional factors, portfolio-aware learning	Leakage, selection bias, turnover, cost, capacity and opacity	Economically constrained, implementation-aware ML
Large language models	Investment information is predominantly unstructured and contextual	Foundation models, embeddings, retrieval and generative extraction	Event parsing, transcript analysis, research assistance and document comparison	Temporal contamination, hallucination, version instability, lack of reproducibility	Grounded, deterministic and auditable systems
Multi-agent AI	The research process is fragmented across specialized functions	Memory, role specialization, debate and workflow orchestration	Research, data, sceptic, execution and risk agents	Correlated errors, feedback loops and weak evaluation protocols. Layered stochastic models can amplify error variance and reduce reproducibility.	Controlled adaptive human–machine organizations

subsequent returns; Banz (1981) documented the size effect; Fama and French (1992, 1993) organized size and book-to-market into a systematic factor framework. The intellectual achievement was not merely the discovery that cheap stocks sometimes outperform. It was the demonstration that valuation could be implemented as a diversified, replicable exposure.

Why value works remains contested. The risk interpretation views cheap firms as exposed to distress, inflexible capital, or adverse states that are not adequately captured by market beta. The behavioral interpretation views investors as extrapolating growth, neglecting unattractive firms, and demanding excessive

compensation for holding companies with disappointing narratives. Lakonishok et al. (1994) argue that contrarian returns are difficult to reconcile with a simple distress-risk account. Fama and French (1992, 1993), by contrast, treat book-to-market as part of a factor structure that organizes average returns. Both interpretations have institutional content. A risk premium can persist because investors dislike the states in which it performs badly; a behavioral premium can persist because delegated investors find the position reputationally or psychologically difficult to hold.

The later history of value revealed how far the academic factor was from a complete strategy. Book equity is a delayed accounting

construct. Its economic meaning varies across banks, manufacturers, commodity producers, and technology firms with an emphasis on intangibles. A mechanically cheap firm may be temporarily mispriced, structurally impaired, or simply measured badly. Industry implementations consequently moved toward composite valuation, sector-relative comparisons, enterprise-value ratios, intangible-capital adjustments, and combinations with profitability or momentum.

The debate over intangibles is especially important. The apparent weakening of book-to-market in knowledge-intensive economies does not prove that valuation ceased to matter. It may indicate that a historically convenient accounting denominator stopped measuring the productive capital that investors care about. The distinction is more than semantic. A factor can decay because arbitrage eliminates its return, because the economic mechanism changes, or because its empirical proxy becomes obsolete. Those explanations imply very different research responses.

2.3 | Momentum: Information Diffusion With a Dangerous Short Leg

Jegadeesh and Titman (1993) established that intermediate-horizon winners tend to outperform losers. Carhart (1997) then incorporated momentum into the standard performance-evaluation toolkit. Momentum altered the intellectual terrain because the information was contained in prices themselves rather than in accounting fundamentals.

A standard score is

$$M_{i,t} = \prod_{k=2}^{12} (1 + r_{i,t-k}) - 1,$$

with the most recent month omitted to reduce contamination from short-term reversal. The simplicity of this score masks several competing mechanisms. Daniel et al. (1998) emphasize investor overconfidence and biased self-attribution. Hong and Stein (1999) model the gradual diffusion of information among traders with differing levels of information. Institutional trading constraints add another channel: analysts, funds, and benchmarked portfolios frequently adjust positions gradually rather than instantaneously.

Momentum's failure mode is not a small reduction in average return but a change in the geometry of the strategy. Daniel and Moskowitz (2016) show that momentum crashes are concentrated after market declines, when volatility is high, and the market rebounds sharply. The loser portfolio then contains distressed, high-beta securities whose prices respond violently to improving conditions. The long-short strategy effectively becomes a short recovery option. Barroso and Santa-Clara (2015) show that momentum volatility is sufficiently predictable to enable volatility scaling to improve risk-adjusted performance in their sample.

The practical lesson is not that momentum should always be volatility managed. Is it possible that an unconditional characteristic portfolio can conceal a conditional state exposure? Once the crash mechanism is recognized, the strategy must be evaluated

alongside its financing, beta asymmetry, short-book liquidity, and response to sudden changes in aggregate risk appetite.

Value and momentum also challenge the idea that factor research produces mutually exclusive theories. Asness et al. (2013) document value and momentum across multiple asset classes and show that the two effects are negatively correlated. Diversification is economically useful precisely because its narratives and failure states differ. A fund that frames the problem as "value or momentum" discards the principal benefit of having both.

2.4 | Quality, Profitability, and Investment

Sloan (1996) documents that the cash-flow and accrual components of earnings have different persistence and that prices do not fully reflect those differences. Cooper et al. (2008) show that asset growth predicts cross-sectional returns. Novy-Marx (2013) establishes gross profitability as a powerful characteristic, while Fama and French (2015) incorporate profitability and investment into a five-factor model.

Quality is attractive to practitioners because it appears to repair naïve value. A cheap firm with robust profitability, conservative financing, and disciplined investment is less likely to be a simple distress trap. Yet "quality" is not one variable. It can refer to gross margins, operating profitability, low accruals, balance-sheet strength, earnings stability, governance, or capital allocation. That flexibility makes the concept useful for portfolio design and vulnerable to specification search. Some deeply distressed or highly idiosyncratic low-quality firms may be better suited to concentrated, case-specific strategies—potentially involving restructuring or other active interventions—than to the diversified, systematic approaches considered in this survey.

The relation between value and quality is therefore more subtle than the phrase "quality at a reasonable price" suggests. Profitability can explain why two similarly valued firms deserve different expected returns. It can also become an expensive defensive style in its own right. Asness et al. (2019) show that high-quality firms command higher prices while quality-minus-junk portfolios have historically earned significant risk-adjusted returns. The institutional trade-off lies in separating durable economics from a valuation premium that has already been capitalized.

2.5 | Low Volatility and Low Beta

The low-risk anomaly challenges the elementary prediction that higher systematic risk should be associated with a higher expected return. Ang et al. (2006) find that stocks with high idiosyncratic volatility have low average returns. Frazzini and Pedersen (2014) show that leverage constraints can flatten the security market line because constrained investors overpay for high-beta assets. Baker et al. (2011) add a delegated-management interpretation: benchmark-relative mandates discourage managers from exploiting low-risk stocks through leverage.

A low-volatility portfolio is therefore not simply a safer version of the market. It is a trade against the institutional demand for

TABLE 2 | Canonical factor families and their institutional interpretation.

Factor family	Economic or behavioral rationale	Typical implementation	Hidden exposures	Failure state	Common institutional adaptation
Value	Distress compensation, extrapolation errors, limits to contrarian arbitrage	Composite valuation scores, often industry relative	Distress, leverage, industries, duration and accounting conventions	Long droughts, structural cheapness and value traps	Profitability conditioning, intangible adjustment, momentum overlays
Momentum	Gradual information diffusion, underreaction, institutional adjustment	Winner–loser ranks, residual momentum, industry-relative momentum	Beta asymmetry, crowded trends, liquidity and turnover	Violent rebounds after market declines	Volatility scaling, beta control, slower horizons and diversified trend measures
Quality	Persistent profitability, earnings reliability, disciplined investment	Profitability, accrual, stability and balance-sheet composites	Growth, duration, defensive sectors and accounting differences	Overpayment for resilient firms; speculative-market reversals	Joint quality–value construction and sector-relative measurement
Low risk	Leverage constraints, benchmark incentives and lottery demand	Minimum variance, low beta, defensive selection	Rates, sectors, financing assumptions and crowding	Underperformance in risk-on rallies; expensive defensiveness	Valuation control, exposure constraints and dynamic leverage
Size	Illiquidity, neglect, limited analyst coverage and distress	Small-minus-big portfolios	Microcaps, delistings, bid–ask spreads and market impact	Gross premium absorbed by implementation costs	Liquidity floors and tradability screens
Investment	Conservative investment associated with higher expected return	Asset-growth and investment sorts	Industry life cycles, financing regimes and corporate maturity	Instability around investment booms	Profitability conditioning and industry-relative signals

high-beta securities. That distinction explains why implementation matters. Minimum-variance portfolios may concentrate in utilities, real estate, consumer staples, or bond-like equities. Betting-against-beta portfolios require leverage and shorting. Both can become expensive when defensive investing attracts large flows.

The low-risk literature is a useful corrective to the habit of describing factor returns as stable properties of characteristics. The anomaly depends on financing constraints, benchmark design, security valuation, and the composition of the investable universe. The portfolio can cease to be attractive even if the underlying leverage constraint does not disappear.

The major factors families are compared in **Table 2**. The table treats a factor as an institutional package: an economic mechanism, a portfolio construction rule, a collection of hidden exposures, and a characteristic failure state.

Figure 3 draws the institutional conclusion from the factor era. A characteristic does not move directly from a paper into a fund. It must survive a sequence of transformations, each of which can invalidate the original evidence.

2.6 | Why Factors Survived Their Own Commoditization

The proliferation of factor products did not eliminate the underlying strategies. It changed where their value resides. A publicly known factor can continue to earn a return when it compensates for an undesirable state, exploits a durable institutional constraint, or requires patience that many investors cannot sustain. What is less likely to persist is an unconstrained return attached to a simple, easily traded public variable.

2.6.1 | The Factor Zoo and the Search for Explanatory Power

The proliferation of reported factors raises an evidentiary problem rather than merely a question of statistical dimensionality. Cochrane's (2011) presidential address helped popularize the term “factor zoo,” while Bartram et al. (2021) note that the cross-sectional stock-return literature had documented more than 450 factors. The relevant question is therefore not simply how many candidate factors have been proposed, but which mechanisms remain economically interpretable, statistically robust, and implementable after costs.

A return predictor becomes fund alpha only after surviving multiple implementation filters.



FIGURE 3 | The institutional alpha lifecycle. [Colour figure can be viewed at wileyonlinelibrary.com]

2.6.2 | Principal Components and Factor Dimensionality

Statistical dimensionality should not be conflated with the number of economically meaningful factors. Kelly et al. (2019) use instrumented principal components to connect firm characteristics with latent factor loadings, illustrating how statistical compression can differ from an economically interpretable factor taxonomy. Principal components describe covariance structure; investable factors are expected to correspond to interpretable and implementable sources of return, some of which may be

conditional on economic states rather than persistent across all environments.

2.6.3 | Replicability, Data Mining, and the Role of Active Investors

Replicability further complicates the interpretation of documented factors. Welch and Goyal (2008), although focused on aggregate equity-premium prediction rather than the cross-section, show how apparently established return-predictive vari-

ables can perform poorly out of sample. In the cross-section, Harvey et al. (2016) and McLean and Pontiff (2015) similarly raise the evidentiary bar through multiple-testing and post-publication evidence. The implication is not that factor research is futile, but that a documented factor must be distinguished from a statistically selected or economically fragile predictor.

Even so, the continued search for factors remains valuable. Factors are one of the main ways in which managers translate economic and financial assumptions into portfolios. Active managers therefore provide a competitive mechanism through which hypotheses are implemented and tested in markets. When those strategies are successful and scalable, their own implementation can compress the returns they seek to exploit.

Factor investing also survived because the published long-short portfolio is not the only economically relevant object. An institution can use factors to improve risk control, diversify fundamental forecasts, hedge unintended exposures, or condition other strategies. A factor can lose its status as a standalone alpha and retain substantial value as a portfolio component.

This point complicates claims that a factor has “died”. A lower long-short mean may reflect genuine arbitrage, changed accounting measurement, expensive factor valuations, altered sector composition, or a rising cost of implementation. A serious diagnosis must distinguish among these possibilities. That requirement leads naturally from factor discovery to portfolio translation.

3 | From Signals to Portfolios: Where Academic and Fund Alpha Diverge

3.1 | The Relevant Optimization Problem

Academic anomaly research commonly asks whether a variable predicts returns after controlling for a benchmark. A fund asks whether the variable improves the feasibility of a portfolio. The institutional problem is closer to

$$\max_{w_t} w_t' \widehat{\mu}_t - \frac{\gamma}{2} w_t' \widehat{\Sigma}_t w_t - TC(w_t - w_{t-1}^+ - 1; A)$$

subject to

$$B_t' w_t = b,$$

$$1 \leq w_t \leq u,$$

$$q_t \in Q_t,$$

$$\|w_t\|_1 \leq L$$

These restrictions can be grouped into four broad classes: benchmark or exposure constraints, position bounds, leverage or gross-exposure limits, and trade-feasibility constraints.

The set Q_t captures the last category, including liquidity, borrow availability, market-participation limits, restricted lists, and order timing. A predictor with a high information coefficient may receive little weight because it duplicates an existing exposure,

relies on unavailable shorts, concentrates in illiquid firms, or changes too rapidly.

This difference is not an indictment of academic abstraction. A simplified long-short portfolio can identify a return pattern more cleanly than a proprietary implementation burdened with many constraints. Problems arise when the evidentiary purpose of the simplified portfolio is confused with proof that an investable strategy exists.

3.2 | Estimation Error is an Economic Cost

DeMiguel et al. (2009) matter beyond portfolio allocation because they reveal a general property of quantitative research: complexity consumes sample information. The more parameters a strategy estimates, the greater the improvement required merely to compensate for the additional estimation error. As model dimensionality grows, finite samples become an increasingly binding constraint—the classic “curse of dimensionality” associated with Bellman (1957). In finance, this problem is amplified by noisy returns, limited point-in-time histories, and the cost of obtaining high-quality data. Leung et al. (2021) show that the statistical gains from machine-learning forecasts translate into more modest economic gains once risk-taking capacity and trading implementation are accounted for. Data quality and availability, therefore, constrain the value of additional model flexibility.

Feng (2026) revisits this issue in the context of factor timing and volatility management. Using recursively implemented portfolios, the paper finds that simple equal allocation remains a demanding benchmark against more elaborate timing and optimization schemes. The contribution is best read as a contemporary robustness check rather than as a universal verdict against factor timing. The sample, factor set, objective function, and cost assumptions remain consequential. Its deeper message is that sophisticated strategies should be compared with strong, simple alternatives, not merely with zero predictability.

This benchmark discipline is often missing in machine-learning research. A neural network may outperform an ordinary least-squares model while failing to outperform ridge regression, a characteristic average, a simple ensemble, or a slower strategy with lower turnover. The relevant incremental contribution is

$$\Delta U = U(\text{complex feasible system}) - U(\text{best simple feasible system}),$$

not the difference between a complex forecast and an intentionally weak baseline.

3.3 | Transaction Costs Belong Inside the Model

Transaction costs are frequently added after a gross strategy has been formed. This practice treats implementation as a haircut on an otherwise well-defined alpha. In many strategies, however, the cost is part of the economic mechanism.

A signal based on gradual information diffusion must trade before the information is fully reflected in prices. Delay changes the

exposure itself. A liquidity-provision strategy earns a return partly because it trades when doing so is costly or uncomfortable. A short-side anomaly may be a compensation for borrow scarcity. Removing the cost from the theory can remove the reason the return exists.

A stylized cost function is

$$TC_{i,t} = s_{i,t} | q_{i,t} | + \eta_{i,t} \sigma_{i,t} \left(\frac{q_{i,t}}{V_{i,t}} \right)^\delta | q_{i,t} | + b_{i,t} | q_{i,t}^- | + o_{i,t}$$

where $s_{i,t}$ is the effective spread, $V_{i,t}$ is available volume, $b_{i,t}$ is the short-borrow cost, and $o_{i,t}$ includes fees and opportunity cost. Almgren and Chriss (2001) formalize the trade-off between market impact and execution risk. Gârleanu and Pedersen (2013) show how predictable returns and trading costs jointly determine dynamic positions. Lesmond, Schill, and Zhou (2004) and Korajczyk and Sadka (2004) demonstrate that momentum's implementability depends on the method used to estimate costs and on the scale of trading. Novy-Marx and Velikov (2016) show that turnover and simple cost-mitigation rules materially alter the net performance of many anomalies.

Frazzini et al. (2012) add unusually valuable institutional evidence by analyzing live trading data from a large money manager. Their estimates imply that many broad anomalies have more capacity than pessimistic cost models suggest, but they also show that cost is security-, strategy-, and scale-dependent. The dispute between optimistic and pessimistic estimates is not a nuisance to be averaged away. It identifies a central empirical uncertainty: the relevant cost curve belongs to a particular investor, execution technology, and trade schedule.

3.4 | Liquidity, Shorting, and Capacity

Amihud (2002) and Pástor and Stambaugh (2003) show that liquidity is associated with expected returns and systematic risk. For a fund, liquidity also determines how quickly a model error can be reversed. A security can be tradable in ordinary conditions and effectively unavailable during a crowded unwind.

Capacity is the amount of capital that can be deployed before marginal net alpha falls below the institution's hurdle:

$$A^* = \sup\{A : \alpha_{net}(A) \geq \alpha_{min}\}.$$

The expression makes the capacity strategy specific. A slower, broad signal can absorb more capital than a short-lived signal concentrated in small firms. A high-Sharpe strategy may have little business value if A^* is small. Conversely, a modest edge can be exceptionally valuable if it is broad, persistent, and cheap to trade.

Shorting introduces another asymmetry. The academic long and short portfolios are normally constructed as if the same securities are equally available on both sides. Live short books face locate failures, recalls, borrow fees, crowded ownership, and asymmetric losses. The short leg of value or momentum may contain precisely the securities for which financing becomes most difficult during stress.

A complementary way to describe the implementation wedge is the Fundamental Law of Active Management. In its original form, Grinold (1989) relates the information ratio to forecast skill and breadth. Clarke et al. (2002) extend the framework to constrained portfolios by introducing the transfer coefficient:

$$IR \approx TC \times IC \times \sqrt{BR}$$

where IC measures forecast skill, BR is the number of independent investment decisions, and TC measures how effectively forecasts are transferred into portfolio positions under constraints. The framework complements the capacity discussion by showing that forecast skill creates value only to the extent that it can be expressed through sufficiently broad and implementable positions.

3.5 | Crowding and Liquidation Half-Life

Khandani and Lo (2011) interpret the August 2007 quantitative-equity losses as consistent with forced deleveraging across related strategies. Lou and Polk (2022) construct comomentum, a measure of abnormal-return correlation among momentum stocks, and link it to arbitrage activity. Their work suggests that crowding is not exhausted by the question of how much volume a portfolio represents today. The relevant issue is how many investors may need to trade in the same direction after a common shock.

A low-frequency strategy may therefore have a short liquidation half-life. Monthly signals do not guarantee monthly risk when margin, volatility targets, or redemption pressure force daily adjustment. The fund's effective horizon is the shortest of those for the signal, financing arrangement, liquidity buffer, and investor capital.

The academic-industry divide is summarized in Table 3. The distinction should not be framed as theory versus reality. Academic research and fund research operate under different contracts. The former values generalization and public verification; the latter values marginal portfolio utility under a specific capital and infrastructure base.

The shift from signal estimation to full-process engineering explains why alpha decay occupies a central place in the evolution of quantitative investing. A signal can be real in the historical sample and still lose its institutional value as knowledge, capital, and implementation technology diffuse.

A useful way to summarise the implementation problem is that a fund's effective investment horizon is often determined not by the signal itself, but by the shortest horizon among several operation constraints:

$$H_{effective} = \min(H_{signal}, H_{financing}, H_{liquidity}, H_{investor})$$

A signal may have a multi-year economic horizon, yet the portfolio can still be forced into daily adjustment by financing constraints, liquidity shocks, or investor redemptions. In practice, the relevant horizon is therefore not the horizon over which alpha is expected to emerge, but the horizon over which capital can remain committed. This helps explain why low-frequency

TABLE 3 | Academic and industry research contracts.

Dimension	Academic research contract	Fund research contract	Main source of divergence
Research outcome	Credible evidence of predictability or model failure	Reliable marginal contribution to net portfolio utility	A statistically significant predictor may add no portfolio value
Data	Reproducible public histories	Point-in-time public, licensed, proprietary and internally generated data	Vendor histories may not reproduce live observations
Benchmark	Accepted factor model or published portfolio	Existing book, capital charge and realistic alternative strategy	Standalone alpha can be redundant
Validation	Out-of-sample evidence, robustness and inference	Walk-forward testing, shadow trading, stress, perturbation and operational rehearsal	Institutional failures frequently lie outside conventional standard errors
Trading costs	Stylized spread or turnover adjustment	Security-, side-, urgency-, venue- and scale-dependent cost model	Costs are nonlinear and state dependent
Capacity	Often secondary	A core determinant of commercial value	Attractive small-stock returns may not support target capital
Interpretability	Economic explanation and transparent empirical design	Diagnosis, accountability, investor communication and kill criteria	Different audiences require different explanations
Reproducibility	External code, definitions and independent replication	Internal lineage, version control, auditability and disaster recovery	Proprietary work can be internally reproducible without being public
Success criterion	New, credible and generalizable knowledge	Repeatable net P&L with acceptable drawdown and operational risk	The objectives overlap but are not identical

strategies can experience high-frequency risk during periods of stress.

4 | Alpha Decay: The Endogenous Life Cycle of Quantitative Advantage

4.1 | Four Forms of Decay

“Alpha decay” is used too loosely. At least four phenomena should be distinguished.

Signal-horizon decay describes the reduction in predictive content as the interval between observation and return realization increases. A textual event may matter for minutes or days; an investment or profitability characteristic may diffuse over quarters.

Calendar-time decay describes a decline in average returns across historical periods. This is the object in post-publication studies.

Scale decay describes the reduction in net return as deployed capital increases, which forces the strategy into weaker opportunities.

Structural decay occurs when the economic behavior being exploited changes. Investors learn, firms adjust disclosures, regulators modify rules, index providers alter composition, or market technology shortens the information-processing interval.

A minimal model is

$$\alpha_{\text{gross}}(t) = \alpha_0 e^{-\lambda t}, \quad h_{1/2} = \frac{\ln 2}{\lambda}.$$

The decay rate λ should not be assumed constant. Publication, vendor distribution, fund flows, market stress, or a new processing technology can change it. Net alpha is

$$\alpha_{\text{net}}(A, t) = \alpha_0 e^{-\lambda t} - C(A, t).$$

This formulation separates a weakening predictor from a rising implementation wedge. The distinction is empirically important because the remedy differs. A weaker predictor requires new information or a revised mechanism. Higher costs may require slower trading, a broader universe, or less capital.

Figure 4 depicts alpha decay as a feedback system. Discovery, publication, arbitrage capital, crowding, and execution costs reinforce one another. **The system is endogenous: successful research changes the market in which the research is evaluated.**

Successful research changes the market in which it is evaluated.

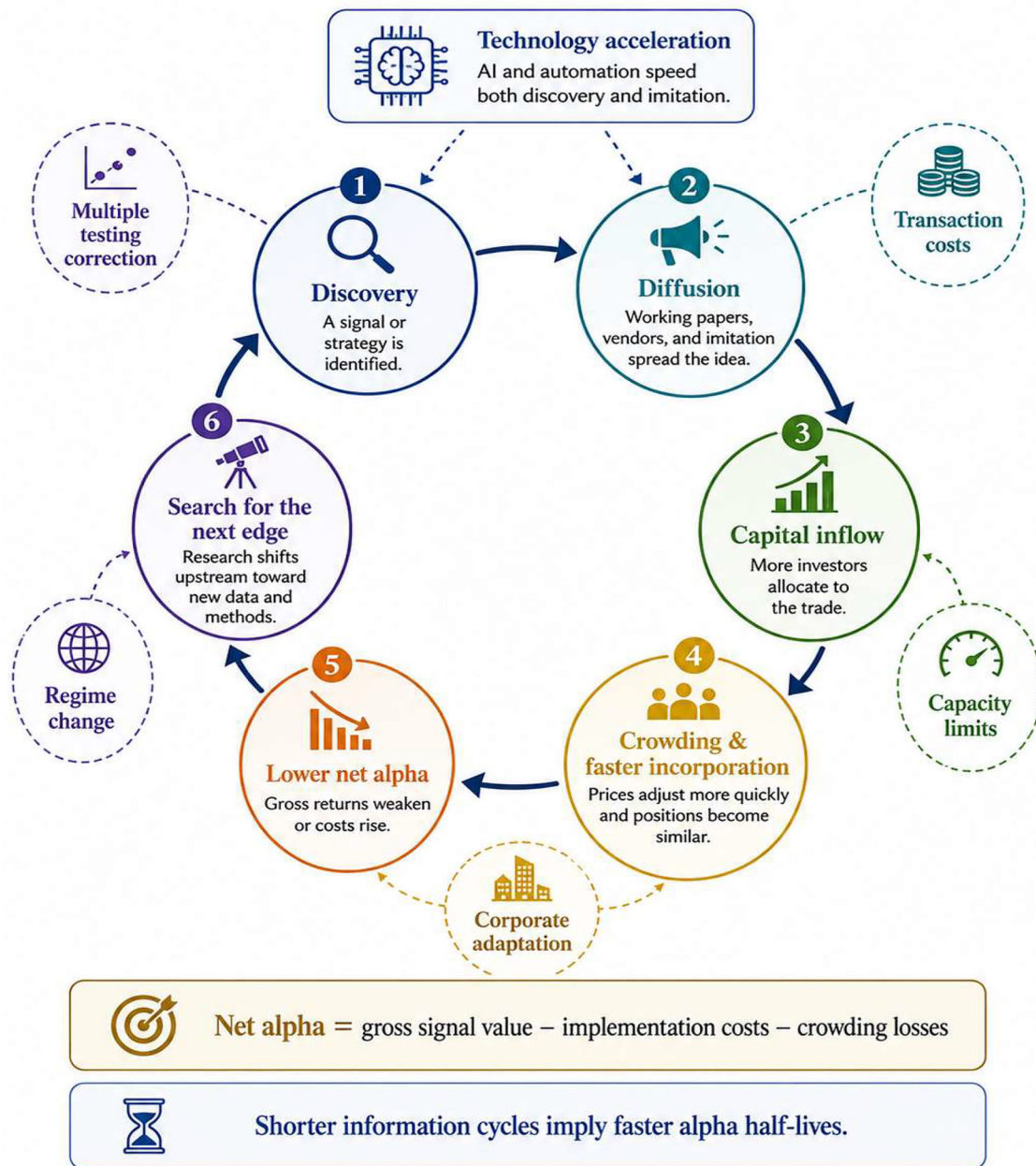


FIGURE 4 | Alpha decay as a feedback system. [Colour figure can be viewed at wileyonlinelibrary.com]

4.2 | Publication, Discovery, and Market Learning

McLean and Pontiff (2015) compare anomaly returns before and after the end of the original sample and after publication. They find substantial attenuation, consistent with a combination of statistical overstatement and investor learning. Chordia et al. (2014) relate declining anomaly profits to greater trading activity and arbitrage capital. These results support the intuitive claim that making a signal easier to recognize and implement reduces its future return.

Yet “publication destroys alpha” is too simple. Jacobs and Müller (2020) examine anomalies across international markets and find

that reliable post-publication decline is concentrated in the United States. If publication mechanically eliminated predictability, similar decay would be expected wherever the paper could be read. Their evidence instead points toward variation in arbitrage capacity, market integration, and implementation feasibility.

The date of publication is itself an imperfect event marker. Ideas circulate in seminars, working papers, practitioner research, and proprietary organizations before journal publication. Conversely, an obscure published result may not attract capital until a later product or replication makes it accessible. The economically relevant variable is diffusion, of which publication is only one observable proxy.

4.3 | False Discoveries are Not Decayed Alphas

Harvey et al. (2016) argue that conventional significance thresholds are inadequate when hundreds of factors have been tested. Feng et al. (2020) develop a formal procedure for evaluating new factors in the presence of a large factor zoo. Hou et al. (2020) apply consistent replication procedures to a broad anomaly set and find that many published effects are weaker or insignificant under common construction and multiple-testing standards. Jensen et al. (2023), however, challenge the strongest replication-crisis interpretation by arguing that theory and Bayesian priors change the meaning of failed stand-alone significance tests.

This disagreement is productive. The replication literature is sometimes presented as a binary contest between “most factors are false” and “most factors are real”. The more useful question concerns the evidentiary status of a characteristic once related signals, prior economic knowledge, publication selection, and portfolio construction are taken into account.

Selection can be represented as

$$\hat{m} = \arg \max_{m \in M} \widehat{SR}_m.$$

Even if each candidate Sharpe ratio is unbiased before selection, the maximum is optimistic. White (2003), Sullivan et al. (2002), and Hansen (2005) develop tests designed to account for data snooping and multiple model comparison. The model set M must include the paths not reported in the final paper: alternative universes, lags, winsorisation rules, holding periods, and portfolio cutoffs.

A weaker post-sample return can therefore mean that investors arbitrated a genuine pattern, that the original estimate selected favorable noise, or that the economic state changed. Only the first is alpha decay in the strict competitive sense. The second is statistical correction. The third is structural instability.

4.4 | Why Decay Appears to be Accelerating

The fixed cost of reproducing quantitative research has fallen. Public code, cloud computing, vendor datasets, standardized backtesting platforms, and pretrained models allow a small team to replicate work that previously required specialized infrastructure. Chen and Zimmermann (2022) make a large collection of cross-sectional predictors available in a harmonized open-source framework. The project improves scientific transparency and simultaneously illustrates why public formulae become less scarce.

Information processing has also accelerated. Machine-readable filings, news feeds, speech-to-text systems, and domain-specific language models shorten the interval between disclosure and systematic interpretation. A signal that persisted because few investors could process thousands of documents may disappear once the task becomes a commodity API call.

More capital now organizes itself around related characteristics. The models need not be identical for trades to overlap. Value

managers can share distressed longs; momentum systems can share recent winners; low-risk portfolios can share defensive sectors; machine-learning systems can converge on similar non-linear rankings because they use the same datasets and loss functions.

Productization contributes another feedback. A characteristic can move from working paper to index, exchange-traded fund, risk model, and portfolio optimizer in a relatively short interval. Product demand can initially strengthen the return as flows move in the same direction, then weaken expected return by raising valuations, and finally amplify liquidation when flows reverse.

Corporate adaptation is less visible but equally important. When investors reward a reported metric, managers change disclosure and operational decisions. A signal becomes a target. The resulting version may retain the label while losing the informational content that originally made it useful.

HFT provides an extreme illustration of why signal-horizon decay should not be confused with the persistence of an alpha-producing system. Individual microstructure signals may decay within milliseconds, yet a trading organization can sustain profitability by continually refreshing signals, improving execution, and maintaining relative advantages in latency or market design. Evidence from HFT therefore shifts the object of persistence from a single forecast to the broader production process. Baron et al. (2019), for example, document persistent differences in HFT-firm performance associated with trading style and relative latency.

4.5 | The Heterogeneous Persistence of Factors

Not all alpha should decay at the same rate. A public accounting transformation is easier to copy than a proprietary execution process. A risk premium tied to undesirable macroeconomic states may persist even when universally known, because knowledge does not eliminate the unwillingness to bear the risk. A behavioral effect can persist when the underlying bias is reinforced by institutional mandates. A microstructure signal may disappear quickly once exchange design changes.

Stambaugh and Yuan (2016) propose mispricing factors that organize a broad set of anomalies, while Kozak et al. (2020) show that a relatively low-dimensional structure spans much of the cross-section. These papers suggest that the factor zoo may contain many empirical manifestations of a smaller number of underlying forces. If so, apparent factor decay can involve migration within a common latent structure rather than complete disappearance.

The practical decomposition is

$$\alpha_t = \alpha_t^{\text{risk}} + \alpha_t^{\text{behaviour}} + \alpha_t^{\text{information}} + \alpha_t^{\text{microstructure}} - \alpha_t^{\text{crowding}}.$$

The components have different half-lives. Public information-processing advantages are likely to decay rapidly. Compensation for bearing disliked states may remain but fluctuate sharply. Institutional effects can persist until mandates, leverage constraints, or investor preferences change. Crowding can turn a slowly decaying average return into a discontinuous loss.

TABLE 4 | Mechanisms and diagnostics of alpha decay.

Mechanism	Observable symptom	Tempting but incomplete interpretation	More informative diagnostic	Fund response
Initial statistical overstatement	Large original result followed by broad out-of-sample weakness	Arbitrage removed the signal	Multiple-testing correction, specification distribution and independent reconstruction	Stronger priors, frozen holdouts and simpler baselines
Publication diffusion	Decline around working-paper circulation or publication	The economic mechanism disappeared	Compare discovery, circulation, publication and product-launch dates	Assume faster diffusion and seek proprietary refinements
Capital crowding	Rising valuation spreads, common holdings and correlated drawdowns	A high recent Sharpe confirms reliability	Holdings overlap, flow sensitivity, financing and unwind stress	Capacity limits and crowding-aware sizing
Transaction-cost inflation	Gross predictability survives while net return falls	The alpha remains intact	Security-level cost and participation-rate sensitivity	Lower turnover, slower execution and broader trading universe
Data commoditization	Vendor coverage expands while edge narrows	More data should strengthen the model	Measure uniqueness of cleaning, timestamps and economic mappings	Compete on transformation rather than raw access
Structural regime change	Return concentrated in historical states	Retraining alone will restore performance	Conditional attribution, break tests and mechanism reassessment	State-aware exposure and model diversification
Corporate adaptation	A reported metric loses information after becoming a target	Accounting relations are permanent	Examine changes in disclosure and managerial behavior	Use harder-to-manipulate and economically linked measures
Technology convergence	Different funds generate correlated forecasts	Better models imply independent alpha	Forecast correlation and trade-overlap analysis	Diversify data, objectives and execution
Financing shock	Abrupt losses across apparently distinct slow signals	Fundamentals changed simultaneously	Margin, borrow, leverage and forced-flow analysis	Liquidity reserves and explicit deleveraging protocols

Table 4 distinguishes the main decay mechanisms. Its diagnostic column is important because the same observed decline can support conflicting narratives.

4.6 | AI and the Red Queen Problem

Artificial intelligence lowers the cost of discovering candidate signals. It can generate code, classify documents, search interactions, and propose hypotheses. The same technology lowers the cost of imitation. A newly described strategy can be reproduced across markets before the original journal article appears.

This creates a Red Queen equilibrium. Research productivity rises while the life of the average public signal shortens. Institutions must improve their discovery and retirement processes merely to preserve the same relative advantage.

The important unit of persistence is therefore not a single factor. It is the organizational capability to form hypotheses, reject false results, implement surviving signals, observe live decay, and

reallocate capital. Alpha is increasingly renewable rather than permanent.

In summary, discovery, publication, arbitrage capital, crowding, and execution costs are not independent stages. They form a self-reinforcing feedback loop in which successful research alters the very market environment on which its success was originally measured. Alpha therefore exists within an endogenous system: once a return pattern becomes known and traded, the process of exploitation changes the future profitability, capacity, and implementation of the strategy itself.

5 | Alternative Data: Information Engineering After the Factor Zoo

5.1 | Alternative Relative to What?

Alternative data is usually described by examples: payment transactions, web traffic, satellite images, app usage, job postings, geolocation, product reviews, patent text, or supply-chain

links. This definition ages badly. Once a dataset becomes widely licensed and incorporated into standard platforms, it may still be called alternative even though access is no longer scarce.

A more useful definition is functional. Alternative data changes the information set through greater timeliness, granularity, coverage, or economic linkage:

$$\mathcal{V}(D) = f(\text{timeliness, granularity, coverage, representativeness, economic mapping}).$$

A dataset can be timely but unrepresentative, granular but economically ambiguous, or predictive in a backfill but unavailable in the same form historically. Goldstein et al. (2021) emphasise that financial big data changes research design, market structure, and the production of economic information. The institutional challenge is not simply to store more observations. It is to preserve their meaning through time.

Figure 5 presents alternative data as a value chain. Raw access is the visible starting point, but the defensible layer often lies later: entity resolution, timestamp control, revision tracking, economic linkage, and execution.

5.2 | Textual Data and Contextual Meaning

Text was the first large alternative dataset to become central to equity research. Tetlock (2007) shows that pessimism in financial media relates to market activity and subsequent returns. Tetlock et al. (2008) link firm-specific news language to earnings and returns. Loughran and McDonald (2011) demonstrate that general-language sentiment dictionaries misclassify many financial terms and develop domain-specific word lists. Da et al. (2011) use Google search volume as a measure of investor attention.

These studies remain important in the LLM era because they establish that representation is economic. “Liability”, “tax”, “capital”, or “risk” do not have stable sentiment outside context. A model that improves linguistic classification can create value only if the resulting context is new to the market and available before the relevant price adjustment.

Corporate language can also reveal relations that standard classifications miss. Hoberg and Phillips (2016) use product descriptions to construct dynamic product-market networks. Hassan et al. (2019) derive firm-level political-risk measures from earnings calls. Li et al. (2021) use machine learning to measure corporate culture from textual communication. These applications are successful because language is not treated as an isolated sentiment score; it is mapped to economically interpretable exposures.

5.3 | Networks and the Propagation of Information

Traditional panel datasets represent firms as independent rows with common time effects. The economy is a network. Cohen and Frazzini (2008) show that information travels gradually through

customer–supplier relationships and that linked firms can exhibit predictable returns.

Network data can capture customer dependence, common technologies, labor-market competition, product overlap, and supply-chain shock propagation. Its attraction is also its danger. A present-day relationship graph can contain links that were unknown or nonexistent in the historical period. Applying a current graph retrospectively creates look-ahead bias even when every individual accounting variable is correctly lagged.

Point-in-time graph construction requires a timestamp for the relationship, not merely for the underlying document. The research system must know when the link was inferred, which evidence supported it, and whether the entity resolution available at the time would have associated the correct firms.

5.4 | Transactions, Attention, and Coverage Drift

Payment and web data offer more frequent information about demand, but the observed panel is often an evolving sample. A card provider gains and loses banks, merchants, regions, and customers. A web platform changes ranking algorithms, device coverage, and bot filters. An apparent growth rate may therefore combine real economic activity with a changing measurement process:

$$\Delta y_{i,t}^{\text{observed}} = \Delta y_{i,t}^{\text{economic}} + \Delta y_{i,t}^{\text{coverage}} + \Delta y_{i,t}^{\text{classification}} + \varepsilon_{i,t}.$$

Time-varying sampling bias is especially dangerous because ordinary firm fixed effects do not remove it. A model can learn the expansion of the vendor rather than the expansion of the company.

The same issue appears in job-posting and app data. Historical rankings may be reconstructed using a current taxonomy; duplicate postings may be removed using algorithms trained on the full sample; firms that disappear may be missing from the archive. Every improvement in the vendor’s historical product creates a question: was this improved history available to the investor at the time?

5.5 | Legal and Organizational Usability

The commercial value of a dataset is constrained by the right to use it. Privacy law, contractual restrictions, material non-public information controls, web-scraping terms, and cross-border data rules can make a predictive source unusable.

Legal review should not occur after a backtest has become attractive. The data acquisition process should document source, license, collection method, permitted use, retention rules, geographic coverage, revisions, entity matching, and dependencies. If a vendor changes methodology or terminates access, the fund needs to know which signals, portfolios, and risk reports are affected.

This governance burden helps explain why public discussions of alternative data often overstate the importance of raw access.

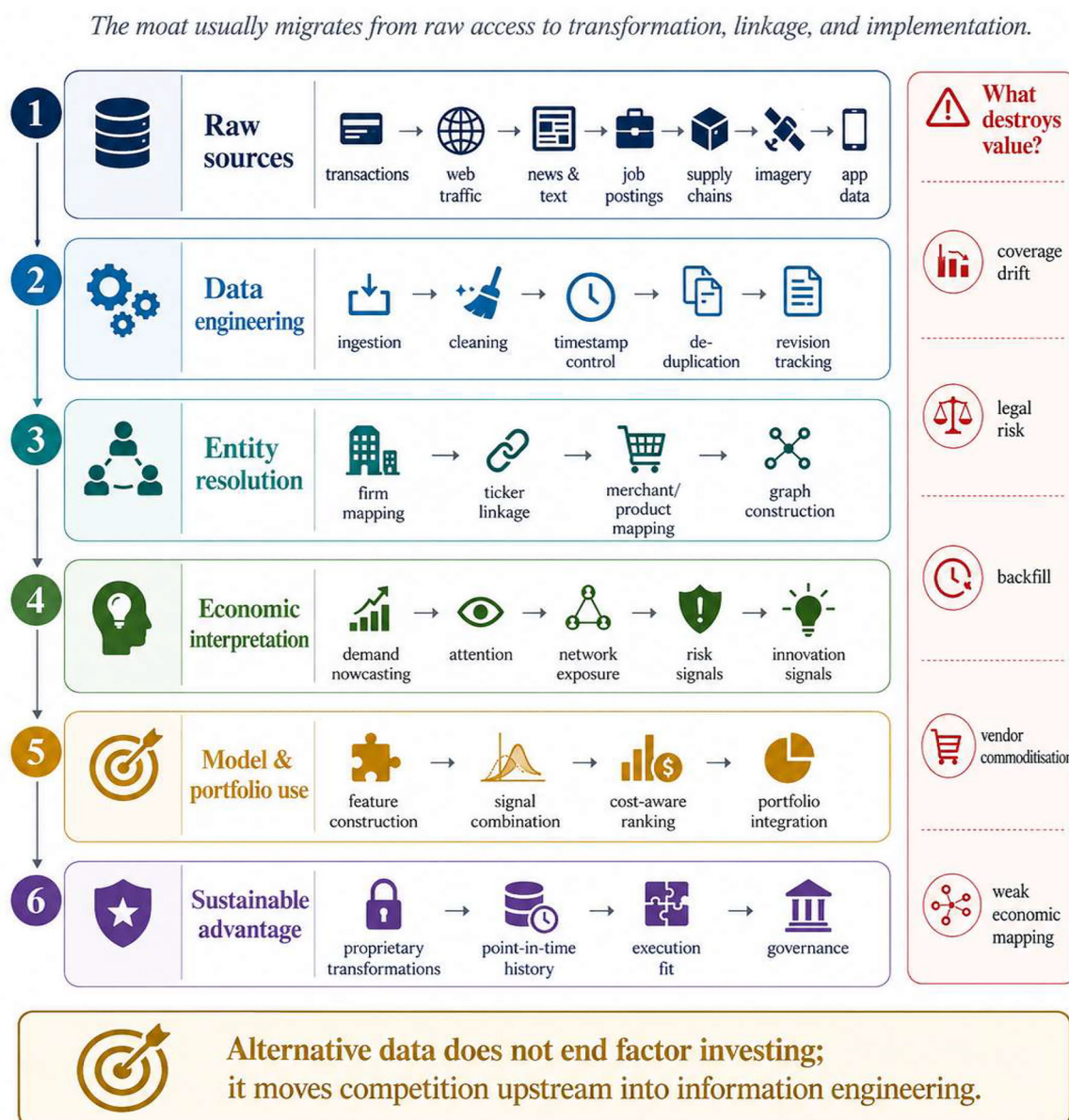


FIGURE 5 | Alternative data: From novelty to durable information. [Colour figure can be viewed at wileyonlinelibrary.com]

Large institutions can purchase the same feed. Their outcomes differ because one has cleaner historical versions, a stronger entity graph, better economic mapping, and an execution process matched to the signal's decay.

The principal data families are compared in Table 5. The table makes clear that the institutional moat is usually not the visible source itself.

ESG data provide a useful stress test of the distinction between a tradable factor and an unstable measurement system. Sustainable preferences can affect prices and expected returns (Pástor et al. 2021), while carbon-intensive firms appear to command a premium consistent with transition risk (Bolton and Kacperczyk 2021). Yet the signal is unusually dependent on data construction. Berg et al. (2022) show that major ESG providers disagree because they differ in scope, measurement, and weighting. Historical scores can also be revised as disclosures, controver-

sies, and estimation methods change. ESG backtests therefore require the same point-in-time provenance, entity matching, and vendor-version controls as other alternative-data strategies. Greenwashing adds a related identification problem: persuasive sustainability narratives may be easier to produce than verifiable changes in emissions or operating practice. As ESG measures enter benchmarks and mandates, their return effects may migrate from information alpha toward crowded preference or regulatory exposures.

5.6 | Commoditization and the Migration of the Moat

Alternative data has its own life cycle. Early access is scarce and technically difficult. Vendors then standardize the source, improve coverage, and create backfilled histories. Research plat-

TABLE 5 | Alternative-data taxonomy.

Data family	Potential information	Principal modelling problem	Characteristic leakage risk	Typical horizon	More defensible moat
Transactions and receipts	Demand, pricing and market share	Representativeness and merchant mapping	Retrospectively reconstructed customer panels	Days to quarters	Stable-panel normalization and company–revenue linkage
Web traffic and search	Attention, demand and product interest	Bots, platform changes and ambiguous intent	Revised histories and present-day classifications	Hours to months	Cross-platform validation and economic interpretation
Corporate text and audio	Guidance, risk, culture and managerial information	Context, boilerplate and speaker attribution	Corrected transcripts and later document versions	Minutes to quarters	Domain representations, document lineage and event ontologies
News and social text	Events, sentiment, controversy and narratives	Duplication, source quality and manipulation	Aggregator time later than original publication	Seconds to weeks	Novelty detection and source-level timestamps
Job postings and labor data	Expansion, skills, location and strategic direction	Duplicate listings and changing platform coverage	Current occupational taxonomy projected backward	Months to years	Skill ontology and coverage-adjusted local panels
Supply-chain and network data	Shock transmission and customer dependence	Missing, indirect and stale relationships	Future-known links in historical graphs	Weeks to quarters	Point-in-time graphs and exposure weighting
Geospatial and imagery	Physical activity, inventories and construction	Weather, image comparability and label quality	Labels trained with future observations	Days to quarters	Proprietary labelled history and physical-domain expertise
App and digital-product data	User growth, engagement and churn	Platform and device selection	Backfilled rankings and survivorship	Days to months	Reconciliation across independent sources
Patent and scientific text	Innovation and technological proximity	Long commercial lags and uncertain product relevance	Later patent families mapped backward	Years	Technical representations linked to products and revenues

forms distribute derived fields. Eventually the dataset becomes a conventional input.

The migration of advantage follows the same path as factor decay. Raw access loses scarcity; cleaning and economic linkage become more important; those transformations eventually diffuse; execution and organizational learning remain.

Alternative data therefore did not replace factors. It changed what a factor could be. A modern characteristic may be an embedding, a graph exposure, an event probability, or a measure of consumer activity rather than a balance-sheet ratio. Once the information set becomes high-dimensional and irregular, machine learning becomes a natural—but not automatically successful—response.

6 | Machine Learning for Alpha Generation: The Implementation Gap

6.1 | Why Machine Learning Appeared

Machine learning entered empirical asset pricing because the conventional information set became too large and interactive for simple additive specifications. A linear model assumes

$$r_{i,t+1} = \beta_0 + \sum_{k=1}^K \beta_k z_{i,k,t} + \varepsilon_{i,t+1}.$$

A flexible model replaces the additive function with

$$r_{i,t+1} = g_{\theta}(\mathbf{z}_{i,t}, \mathbf{m}_t) + \varepsilon_{i,t+1}.$$

The economic case for flexibility is strong. Valuation can matter differently for profitable and unprofitable firms. Momentum risk

changes with market conditions. Liquidity affects the value and cost of other signals. Missingness may itself be informative. Interactions are not a statistical luxury; they can represent the conditional nature of expected returns.

Gu et al. (2020) provide the benchmark equity evidence. Their comparison of linear models, trees, and neural networks shows that nonlinear methods improve out-of-sample return prediction and portfolio performance, with momentum, liquidity, volatility, and valuation among the important inputs. Freyberger et al. (2020) estimate characteristic effects nonparametrically and find substantial sparsity and nonlinearity. Kelly et al. (2019) use instrumented principal components to connect characteristics with latent factor loadings. Gu, Kelly, and Xiu (2021) use a conditional autoencoder to impose an asset-pricing structure on nonlinear latent factors. Chen et al. (2024) estimate deep stochastic discount factors using no-arbitrage restrictions. Leipold et al. (2022) show that the relative performance and predictor importance of machine-learning models vary in the Chinese market. Bagnara (2024) surveys this literature and emphasizes both its predictive promise and its exposure to overfitting and interpretability problems.

These papers establish that machine learning is a serious empirical method. They do not establish that most machine-learning studies produce tradable alpha.

6.2 | The “95 Per Cent” Claim

Practitioners frequently say that almost all machine-learning research fails to enter production. The exact figure—often stated as 90 or 95 per cent—has no defensible census behind it. It should not appear in an academic paper as an empirical estimate.

The intuition behind the claim is nevertheless sound. Production requires a conjunction of conditions, and attrition is multiplicative. If a project has probability p_j of surviving implementation stage j , then

$$P(\text{production}) = \prod_{j=1}^J p_j.$$

Moderately demanding filters can reduce a large research pipeline to a small number of deployable systems without any one stage accounting for the majority of failures.

Figure 6 depicts this funnel. Its numbers are illustrative and should not be interpreted as a measurement of the literature.

The economically important point is that the filters are not independent in practice. Leakage often produces unstable turnover; unstable turnover magnifies cost-model errors; cost errors reduce capacity; low capacity encourages concentration; concentration raises regime sensitivity. The apparent “model problem” is frequently a system problem.

6.3 | Leakage: The Foundational Failure

Leakage occurs when a model uses information unavailable at the intended decision time. The error can enter through reporting dates, revisions, universe selection, labels, preprocessing, text, or graphs.

A point-in-time variable should be indexed by its historical vintage:

$$x_{i,t}^{(v)} = \text{value of feature } i \text{ observable in vintage } v \leq t.$$

The chronological ordering must satisfy

$$t_{\text{event}} \leq t_{\text{release}} \leq t_{\text{ingest}} \leq t_{\text{feature}} \leq t_{\text{decision}} \leq t_{\text{trade}}.$$

An accounting observation associated with December cannot be used in January if the statement was released in March. A present-day index membership list cannot define the historical universe. A corrected transcript cannot replace the original document. A supply-chain graph cannot include relationships discovered later.

Random train–test splits are especially hazardous in panels. Adjacent observations from the same company and market state appear in both samples, allowing the model to exploit persistence that would not be available across time. Bergmeir et al. (2018) discuss cross-validation in dependent settings; financial applications require chronological folds, purging around overlapping labels, and embargoes where information windows intersect.

Text models introduce a deeper form of contamination. A pre-trained language model may have encountered future outcomes during training even when the prompt contains only historical documents. A historical LLM backtest therefore requires a model whose training cutoff, retrieval corpus, prompt, and version are fixed. Otherwise “point in time” applies to the input but not to the representation.

6.4 | Overfitting is a Property of the Research Organization

Overfitting is often reduced to the ratio of parameters to observations. In quantitative research, the relevant model includes every choice made by the research process:

$$\hat{\mathcal{M}} = \arg \max_{\mathcal{M} \in \mathbb{M}} \hat{U}_{\text{validation}}(\mathcal{M}),$$

where $\mathbb{M}$ includes target definitions, universes, lags, missing-data rules, transformations, architectures, hyperparameters, portfolio cutoffs, neutrality constraints, and cost assumptions.

A final neural network can be regularized while the organization remains severely overfit because dozens of researchers inspected related results and abandoned unsuccessful paths. The unpublished search tree is part of the effective model.

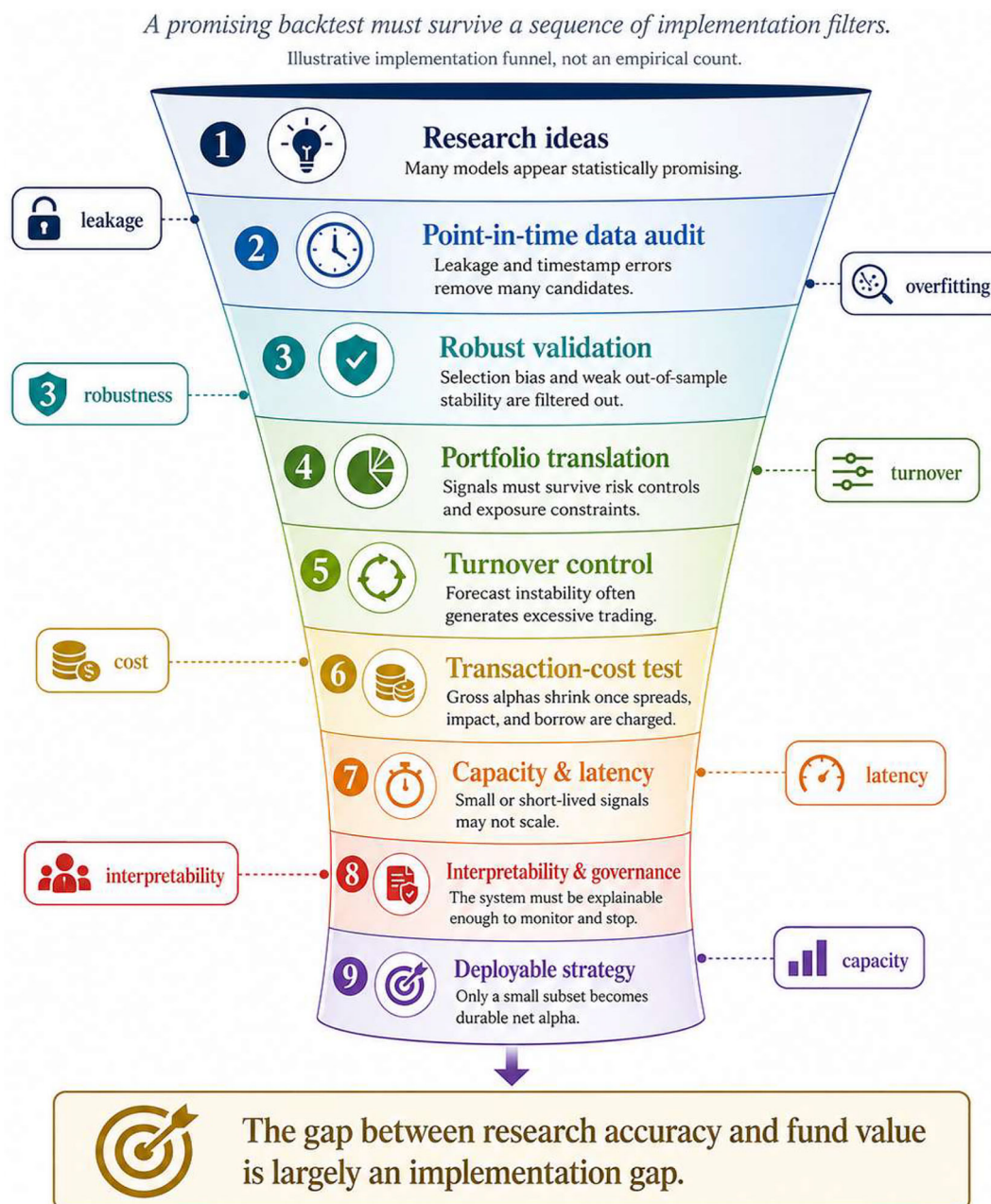


FIGURE 6 | Why most machine-learning results do not become fund P&L. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12178)]

The panel's row count exaggerates the effective sample. Millions of security-month observations share macroeconomic states, industries, flows, and common shocks. Forty years of data contain only a handful of inflation regimes, liquidity crises, momentum crashes, or technological transitions. Cross-sectional breadth helps estimate relative responses but does not create new histories of rare regimes.

The proper response is not endless robustness. A strategy can be varied until some specification survives every requested test. **Better controls restrict researcher freedom before results are known: experiment registries, frozen holdouts, nested validation, strong simple baselines, independent reconstruction, and shadow trading.**

6.5 | Predictive Accuracy is Not Portfolio Utility

Machine-learning studies often optimize mean-squared error, classification accuracy, or rank correlation. Funds trade portfolios. A model can improve average prediction while degrading the economically relevant ranking.

The cross-sectional information coefficient is

$$IC_t = \text{corr}(\text{rank}(s_{i,t}), \text{rank}(r_{i,t+1})).$$

It does not reveal whether the signal lies in liquid or illiquid firms, whether the long and short sides are symmetric, whether the

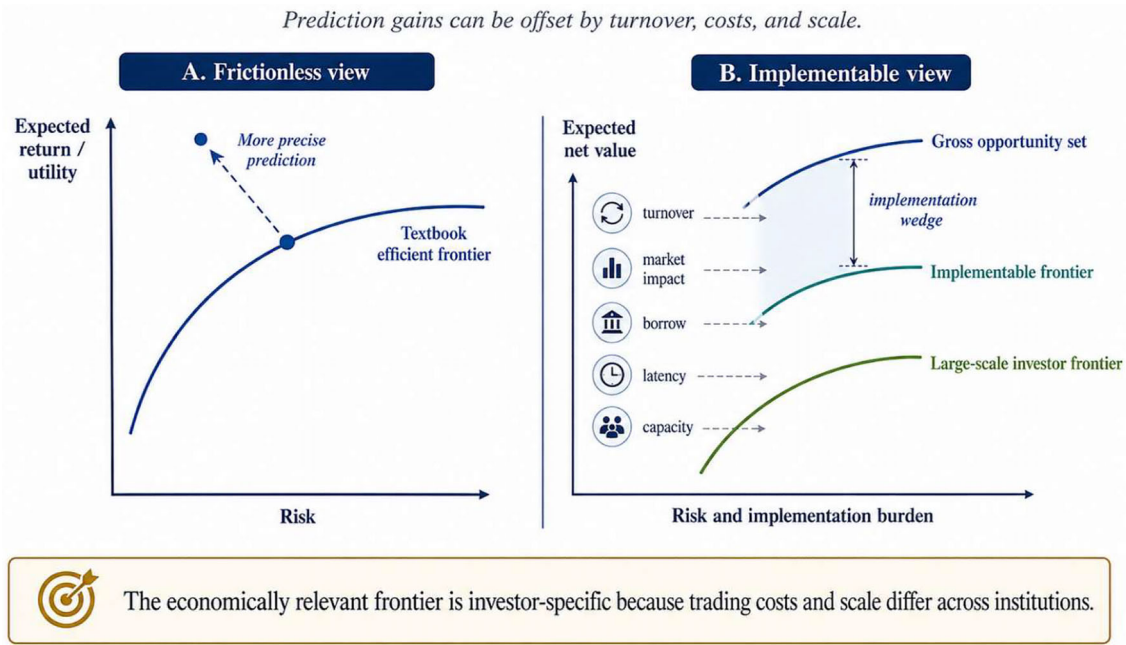


FIGURE 7 | From the textbook frontier to the implementable frontier. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12128)]

forecast is redundant with the existing book, or how rapidly the ranks change.

A portfolio-aware objective is

$$\theta^* = \arg \max_{\theta} \mathbb{E}[\mathbf{w}_t(\theta)' \mathbf{r}_{t+1} - \frac{\gamma}{2} \mathbf{w}_t(\theta)' \Sigma_t \mathbf{w}_t(\theta) - TC_t(\Delta \mathbf{w}_t(\theta))].$$

This does not imply that every model should directly maximise estimated Sharpe ratio. Sharpe-maximising neural networks can overfit the portfolio objective more severely than a supervised model. It implies that evaluation should follow the forecast through the actual position and trade.

Jensen et al. (2026) make this point directly. A high-performing frictionless machine-learning portfolio can generate turnover so extreme that its net frontier is inferior to a model trained to internalize implementation. **Figure 7** illustrates the wedge.

6.6 | Turnover as Forecast Instability

Turnover is not merely an optimizer output. It is the derivative of the forecast process.

$$TO_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}^+|.$$

Scores change because information changes, but also because noisy features fluctuate, trees cross split thresholds, model parameters are retrained, or constraints interact discontinuously. Two models with similar average accuracy can have radically different turnover.

A useful decomposition separates informational turnover from estimation turnover. The former reflects changing expected

returns and may deserve to be traded. The latter reflects model noise and is a pure implementation loss.

Partial adjustment provides a simple control:

$$\mathbf{w}_t = (1 - \rho_t) \mathbf{w}_{t-1}^+ + \rho_t \mathbf{w}_t^*,$$

where ρ_t depends on signal confidence, cost, liquidity, and expected decay. Buffer zones, score smoothing, no-trade regions, and turnover penalties serve a similar purpose. Their value is not cosmetic cost reduction. They force the researcher to identify how much of the forecast is strong enough to justify a trade.

6.7 | Cost-Model Risk

A fixed cost per unit of turnover cannot represent the costs faced by a large fund. Spread, impact, borrow, and opportunity cost vary by security, side, urgency, venue, market state, and crowding.

Machine-learning systems can exploit errors in the estimated cost model. If the optimiser is given a smooth impact curve that understates cost in certain names, the system will concentrate precisely where the model is most optimistic. The cost function becomes another learned surface, and its misspecification becomes another source of alpha illusion.

A robust strategy should be evaluated under base, stressed, crowded, and liquidation cost scenarios. Positive net return under one estimated curve is not enough. The return should remain acceptable when cost parameters are perturbed in economically plausible directions.

6.8 | Interpretability as Institutional Functionality

The standard debate contrasts interpretable linear models with opaque neural networks. The distinction is inadequate. A linear forecast embedded in thousands of portfolio constraints can be operationally opaque. A complex model can be institutionally understandable if its exposures, marginal trades, scenario responses, data lineage, and failure boundaries are observable.

Interpretability has several purposes. The researcher needs error diagnosis. The portfolio manager needs to understand why positions changed. Risk management needs hidden concentrations and stress sensitivities. Senior governance needs limits and stop criteria. Clients and regulators may require a coherent account of the process.

Post-hoc feature importance does not meet these needs. Rudin (2019) warns against using explanations of black-box models where interpretable models are feasible. In investment applications, an explanation should be tied to decisions: which data release changed the forecast, which securities contribute to risk and expected return, how the trade responds to counterfactual inputs, and whether the current observation lies outside the training distribution.

Economic constraints can improve both interpretation and robustness. Monotonicity, economically meaningful feature groups, residualization, hierarchical structures, no-arbitrage conditions, and explicit factor attribution limit implausible behavior without forcing every component to be linear.

6.9 | Latency and Organizational Time

Latency matters at every investment horizon:

$$\tau_{\text{total}} = \tau_{\text{release}} + \tau_{\text{vendor}} + \tau_{\text{processing}} + \tau_{\text{model}} + \tau_{\text{decision}} + \tau_{\text{execution}}.$$

If alpha decays at rate λ ,

$$\alpha_{\text{exec}} = \alpha_0 e^{-\lambda \tau_{\text{total}}}.$$

A news classifier can be statistically impressive and economically useless if the market reacts before the fund can ingest, validate, approve, and trade the result.

The relevant latency is often organizational. Data procurement, legal approval, model review, portfolio integration, and production engineering may take months. By deployment, a vendor's coverage or the market's use of the data may have changed. A model that is cheap to retrain but expensive to approve is not genuinely adaptive.

6.10 | Capacity and the Business Value of Prediction

Capacity should be reported as a curve:

$$\alpha_{\text{net}}(A), SR_{\text{net}}(A), \text{Drawdown}(A),$$

rather than as a single cost-adjusted return.

Machine learning can reduce capacity by concentrating on the strongest predicted firms, repeatedly selecting unusual small stocks, relying on short-lived information, or converging with competitors trained on the same sources. A strategy with a Sharpe ratio of three and capacity of £5 million can be less valuable to a large fund than a broad strategy with a Sharpe ratio of one.

The failure modes and remedies are assembled in Table 6. The deployment criterion is intentionally stricter than a positive backtest.

6.11 | What Has Worked

The strongest evidence supports a narrower view of successful financial machine learning than popular discussion implies.

Nonlinear cross-sectional ranking works when interactions are economically plausible and the validation is disciplined. Regularization works because many characteristics are redundant and noisy. Latent-factor models work when machine learning is used to estimate conditional exposures under an asset-pricing structure. Deep learning is most convincing when the representation problem is genuinely complex and the architecture incorporates no-arbitrage or other economic restrictions. China illustrates why cross-market transfer is an economic, not merely statistical, problem. Leippold et al. (2022) find that liquidity is the dominant machine-learning predictor in Chinese equities; predictability is also pronounced among small stocks in a retail-heavy market. Transaction costs and capacity are therefore central. Factor definitions require local adjustment: Liu et al. (2019) show that the smallest firms contain substantial “shell value,” so conventional size and value sorts can mix expected-return effects with the option value of a reverse merger. Implementation is further shaped by designated-stock shorting eligibility and regulatory intervention. Evidence from the 2010 pilot indicates that relaxing short-sale and margin-trading restrictions improved price efficiency for eligible securities (Chang et al. 2014). Alpha decay should consequently be market-specific: slower incorporation may preserve gross predictability, while shorting limits, price limits, retail flows, and high trading costs determine whether it is monetizable.

Implementation-aware learning may become the most important category. Its aim is not to maximize forecast accuracy and subtract costs later, but to learn a portfolio or target that recognizes turnover and impact during model construction. Jensen et al. (2026) exemplify this approach.

The least convincing applications take an unconstrained model, report a high long-short Sharpe ratio, and interpret the output through feature-importance plots. That workflow leaves the central fund questions unanswered.

The broader conclusion is not that simple models always win. It is that **complexity should be spent where the economic object demands it**: interactions, text, images, graphs, nonlinear costs, and conditional exposures. **Complexity is poorly spent when it only enlarges the number of ways to select a favorable history.**

TABLE 6 | Machine-learning failure modes and fund remedies.

Failure mode	How it enters research	Why ordinary out-of-sample testing may miss it	Institutional control	Deployment criterion
Leakage	Final databases, random splits, future-known universes and contaminated text	Train and test sets inherit the same historical error	Point-in-time vintages, purged chronological folds and lineage audit	Every feature has verifiable availability and processing times
Research overfitting	Repeated model, feature and portfolio selection	The final test set becomes indirectly selected	Experiment registry, nested validation and independent reconstruction	Performance survives frozen data, alternative pipelines and strong simple baselines
Turnover instability	Noisy scores and discontinuous ranks	Prediction metrics ignore trading	Buffers, partial adjustment, turnover penalties and trade attribution	Expected alpha per marginal trade remains attractive
Cost misspecification	Constant basis-point haircut or optimistic impact curve	Average historical costs omit side, urgency and scale	Security-level spread, impact and borrow models with stress scenarios	Positive value under stressed and liquidation costs
Low capacity	Concentration in illiquid or short-lived opportunities	Equal-weight backtests do not represent capital deployment	Participation limits and full capacity curves	Strategy clears its hurdle at target capital
Latency mismatch	Historical simulation assumes immediate processing	Backtest trades at release rather than achievable execution time	Full release-to-trade timestamp reconstruction	Production latency is short relative to signal half-life
Weak interpretability	Post-hoc explanations of unstable mechanisms	Aggregate return conceals changing exposures	Decision-level attribution, counterfactuals and model-version lineage	Trades can be challenged, diagnosed and reversed
Regime fragility	Model learns the dominant historical state	Average test statistics pool different regimes	Conditional attribution, break tests and state stress	Exposure limits are specified for known weak states
Pipeline fragility	Missing fields, vendor changes and software drift	Historical data are clean and complete	Shadow systems, anomaly alarms, failover and versioning	Safe behavior under degraded inputs
Governance failure	No named owner, escalation path or kill rule	Governance does not appear in return statistics	Approval, permissions, audit logs and stop procedures	Accountable ownership and deterministic termination criteria

7 | Large Language Models and Multi-Agent AI: From Forecasting Tools to Research Infrastructure

7.1 | What LLMs Change

Financial natural-language processing previously required task-specific dictionaries, labelled datasets, and specialized classifiers. Large language models supply a general contextual representation that can be adapted through prompting, embeddings, retrieval, or fine-tuning.

Their most important financial role is not conversational. It is representational:

document → *contextual representation* →
structured event or feature → *quantitative decision*.

BloombergGPT demonstrates the feasibility of a large model trained on a mixture of financial and general text. Chen et al. (2022) study the relation between language-model representations and expected returns. Kirtac and Germano (2024) examine sentiment trading with LLMs. Lopez-Lira and Tang (2026) document return predictability from LLM interpretation of news headlines and explicitly consider the implications of broader AI adoption for information diffusion.

The evidence does not imply that an LLM understands asset pricing in the same sense as an economist or portfolio manager. It implies that contextual language representations can extract distinctions that simpler dictionaries miss. There are several issues needed to be considered when integrating AI into asset management practices:

7.1.1 | Immediate Applications of AI in Active Management

This is a direct and immediate use case for AI, but it is only one of many assistive applications that could affect the asset-management industry. When a task is already handled reliably by a deterministic and auditable process, replacing it with a probabilistic LLM may add complexity without adding economic value. AI is likely to be most useful where it accelerates model development, implementation, coding, or unstructured-information processing while preserving well-defined inputs and validation.

7.1.2 | Limits and Risk Controls

The case for autonomous AI-driven trading is less straightforward. Allowing an AI model to trade independently may require costly guardrails, particularly when deterministic alternatives can deliver more controlled and auditable outcomes. Even so, AI can reduce the development and implementation costs of similar strategies. Wherever AI lowers operational frictions—and thereby narrows the gap between gross and net alpha—the industry is likely to adopt it.

7.1.3 | Cost Uncertainty

A key uncertainty is the future cost of AI models. If the most capable models remain expensive and their investment value depends on proprietary integration, specialized capabilities may remain concentrated among a smaller group of firms. If capable models and infrastructure become broadly affordable, some current scale advantages could narrow.

AI integration into active management could therefore follow two contrasting paths. In one scenario, adoption becomes widespread and costs are broadly similar across managers, allowing smaller firms to access increasingly comparable tools and infrastructure. At the other extreme, the most capable models and the proprietary systems required to integrate them could remain concentrated in large money-management platforms. In that case, AI would reinforce existing scale advantages. Regardless of which path dominates, adoption is likely to proceed unevenly across use cases and institutions.

Google's AI & Economy ATLAS (Activity, Task, Landscape, and Adoption Study) provides useful early evidence on how AI is being used in practice. Its first dataset is built from 15 million aggregated and de-identified interactions across the Gemini App, AI Mode, and the Gemini API, spanning more than 150 countries, 140 languages, 800 occupations, and 4000 tasks. Workplace use is broad but shallow: in a typical occupation, AI is used for about 21% of tasks, and most workplace interactions are collaborative or assistive rather than fully automated; fewer than 10% fully automate a task (Ischenko and Strand 2026). The evidence is therefore more consistent with augmentation than with near-term autonomous replacement, although the study does not isolate asset management specifically.

Figure 8 places LLMs inside a controlled quantitative stack. They transform documents, assist retrieval, and generate structured

candidates. Timestamping, risk, portfolio construction, cost, and execution remain deterministic or separately validated.

7.2 | Why LLMs are Not Price Oracles

An LLM is trained to model language. It is not trained by default to preserve data vintages, satisfy no-arbitrage, minimize market impact, respect a position limit, or distinguish a plausible explanation from the actual cause of a return.

Hallucination is the most visible problem but not necessarily the most dangerous. A false company fact can be checked. Temporal contamination can be much harder to detect. A model pretrained on later news may implicitly know the outcome of a historical event. A backtest using historical prompts can therefore contain future information even when the supplied text is clean. A subtler failure mode is a plausible but imprecise output: because LLMs generate fluent and context-sensitive explanations, errors can be difficult to detect, particularly in long responses. For example, an LLM may accelerate the collection of EBITDA figures while failing to preserve differences in firm-, sector-, or jurisdiction-specific definitions and adjustments. If those distinctions are silently normalized, the resulting dataset may appear coherent while embedding a systematic measurement error.

Prompt instability creates another problem. Small changes in instructions, demonstrations, sampling temperature, or provider model can alter classifications. If an output cannot be reproduced under a frozen version, the signal cannot be audited.

Narrative overfitting is particularly relevant to finance. Language models are excellent at producing coherent causal stories. Markets also invite stories because every price movement can be rationalized after the event. A model that explains a winning backtest may be generating a persuasive account of a selected pattern rather than identifying a stable mechanism.

The proper control is not to prohibit generative models. It is to restrict their authority, containing outputs and analyses within well-established guidelines. Outputs should be linked to source passages, converted into defined schemas, timestamped, versioned, and passed through conventional tests. The LLM may propose that guidance weakened; a deterministic system should identify the document, sentence, release time, prior comparison, numerical exposure, and portfolio consequence. The appropriate oversight model—human-in-the-loop (HITL) or human-on-the-loop (HOTL)—depends on the use case and on the authority delegated to the system. Guardrails, escalation criteria, source traceability, and audit trails should therefore be designed at the use-case level rather than added after deployment. There are some important concerns on AI control:

7.2.1 | Limits of AI-Based Auditing

Independent AI systems may assist in auditing one another, but this does not guarantee error cancellation. Correlated model assumptions, shared training information, or interacting stochastic errors can amplify uncertainty rather than reduce it. The next two sections discuss these issues in greater detail.

*Language models are most useful as controlled representation and research tools,
not as unrestricted trading oracles.*

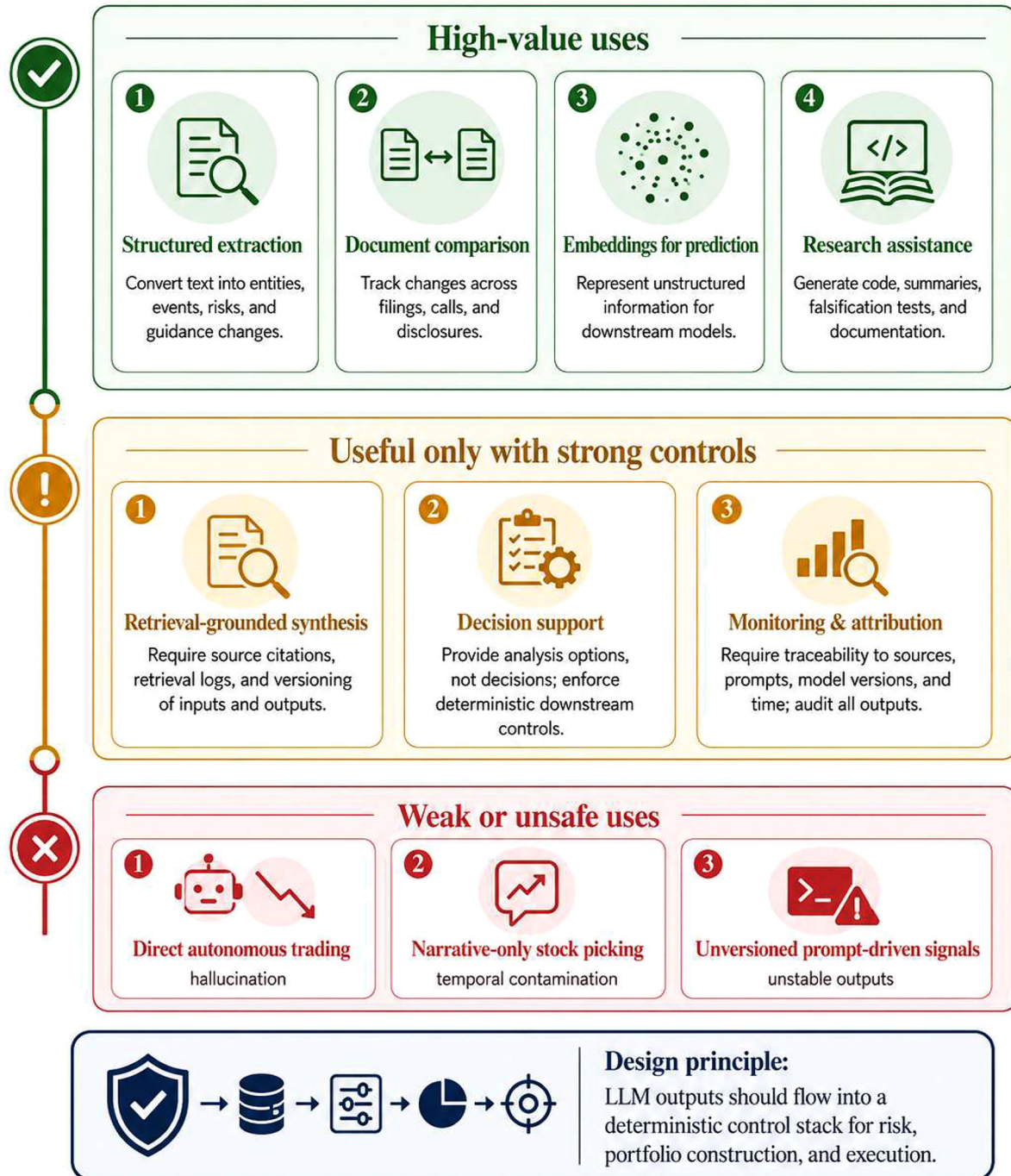


FIGURE 8 | Where LLMs add value—and where they do not. [Colour figure can be viewed at wileyonlinelibrary.com]

7.2.2 | Risk Management in an AI-Driven Environment

The central conclusion is that risk management must evolve in response to AI. The core tools and the distinction between the first, second, and third lines of defense may remain broadly the same. However, governance must become more customized if managers are to reap AI-generated alpha—often less persistent by nature—while also preventing costly model-driven errors.

7.2.3 | Validation Challenges

Large language models are opaque and complex, making them difficult to validate through traditional model-review or attribution frameworks. Some models now provide data sources and references, but those references can still be incomplete, misleading, or incorrect. Similarly, tools that measure response variance across prompts, including Python packages designed for this purpose, are useful but only begin to address the problem.

7.2.4 | Preventive and Detective Controls

Accepting the black-box characteristics of AI requires both preventive and detective controls. **Preventive measures** include avoiding AI in cases where deterministic alternatives are more reliable, auditable, or cost-effective. **Detective measures** are equally important: humans should review outputs, committees should approve and monitor use cases for drift, and organizations should reassess new models and versions as the technology evolves. Vendors and their AI policies should also be questioned, documented, and monitored over time.

7.3 | Near-Term Use Cases With a Credible Evidentiary Path

Structured extraction is the clearest application. The model identifies entities, events, guidance changes, risk disclosures, products, or locations. The output can be compared with labelled data and traced to the source.

Retrieval and evidence synthesis are useful when the corpus is controlled. A model can compare several years of filings, assemble passages on a specific risk, or summarize supplier exposures. The quality of the result depends less on eloquent prose than on retrieval recall, document timestamps, and claim-level traceability.

Embeddings allow language representation to enter a conventional prediction model. This separates the representation task from portfolio construction and makes incremental value easier to measure.

Research assistance may create more value through rejection than discovery. Generating code, checking definitions, proposing falsification tests, or identifying related literature can reduce the cost of ruling out weak ideas. A system that catches a timestamp error can be economically more valuable than one that proposes another marginal factor.

The least defensible use is **unconstrained end-to-end stock selection**. It combines opaque training data, unstable inference, missing cost models, ambiguous timing, and no clear accountability.

7.4 | Multi-Agent Systems as an Organizational Metaphor

A multi-agent system assigns specialized roles to interacting models. A data agent checks lineage and anomalies. A research agent proposes tests. A fundamental agent interprets statements. A sceptic searches for leakage and counterexamples. A risk agent examines exposures. An execution agent estimates cost and capacity. A portfolio layer integrates only those outputs that satisfy deterministic constraints.

The architecture is attractive because real funds are already multi-agent systems. Portfolio managers, researchers, engineers, traders, risk teams, lawyers, and investment committees process different information under different incentives. Arti-

cial agents attempt to replicate or augment that division of labor.

FinCon uses a manager–analyst hierarchy and verbal reinforcement. FinMem combines profiling, layered memory, and decision modules. TradingAgents assigns specialized research, trading, and risk roles. These systems are real research artefacts with identifiable authors, rather than the placeholder citations that appeared in the earlier draft. Yu et al. (2025) subsequently published FinMem in IEEE Transactions on Big Data.

Figure 9 presents the architecture as an operating model rather than a claim of autonomous fund readiness.

7.5 | Correlated Agents and False Diversity

Multiple agents do not automatically provide independent judgement. If all use the same foundation model, retrieval corpus, prompt conventions, and objective, their errors can be highly correlated.

Let e_m denote the error of agent m . The variance of a combined recommendation is

$$\text{Var}\left(\sum_{m=1}^M \omega_m e_m\right) = \sum_{m=1}^M \sum_{n=1}^M \omega_m \omega_n \text{cov}(e_m, e_n).$$

Adding agents with similar errors may increase confidence without improving accuracy. Debate can reinforce a shared false premise if every participant retrieves the same contaminated document.

Genuine diversity requires different information, objectives, model families, or verification tools. A sceptic agent should not be instructed to produce a politely balanced view. Its task should be to invalidate the proposal through timestamp checks, alternative definitions, known factor exposures, cost stress, and contradictory evidence.

7.6 | Memory and Self-Confirmation

Memory is central to agentic systems because it allows the system to retain past observations, analyses, and outcomes. It also creates a new form of contamination. Context management creates a trade-off. Freezing prompts and retrieval context improves comparability across runs, whereas excessively long or repeatedly extended contexts can accumulate stale assumptions and self-reinforcing errors. The appropriate degree of context persistence therefore depends on the use case: exploratory research may tolerate controlled variation, while production research requires stricter versioning, resets, and reproducibility checks.

An agent may retrieve its own previous conclusion as if it were external evidence. A post-hoc explanation of a failed trade can enter memory and later be treated as a fact. Winning trades can receive richer narratives than losing trades, producing a machine version of self-attribution bias.

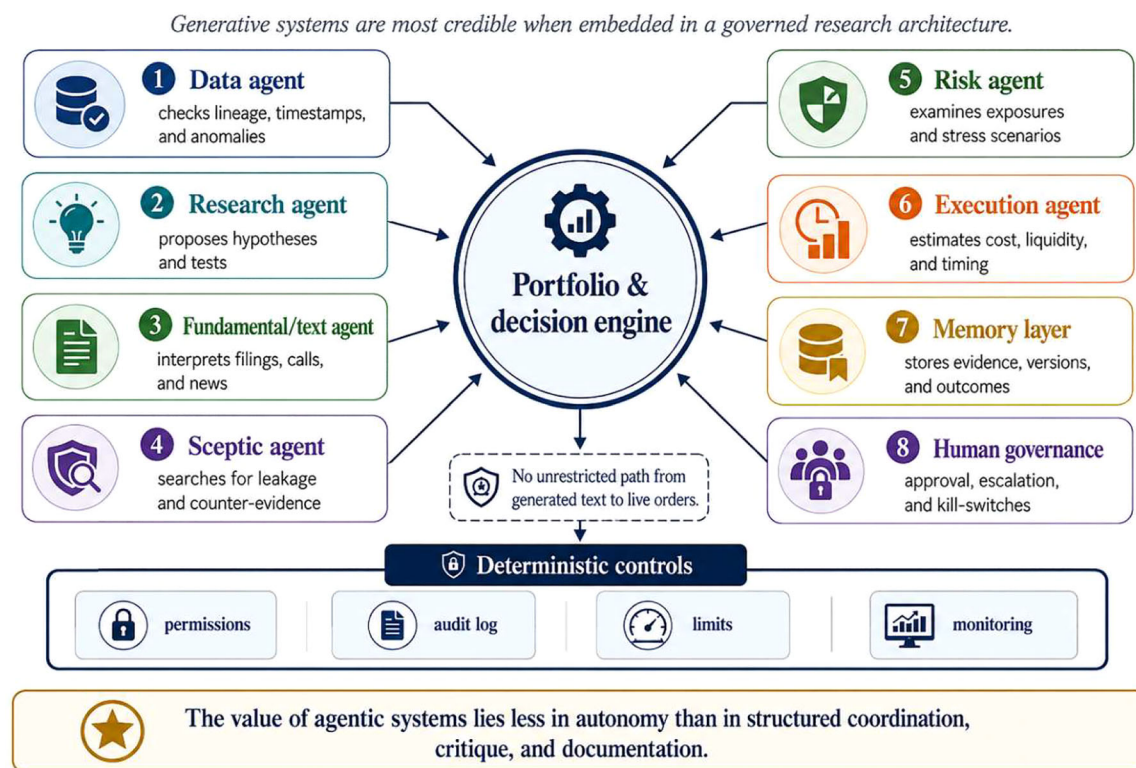


FIGURE 9 | Multi-agent quant research as an operating model. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12026)]

Institutional memory should therefore distinguish raw evidence, model inference, human judgement, decision, and realized outcome. Each stored object needs provenance:

$$m_j = (\text{source, source time, ingest time, model version, claim type, confidence, decision link})$$

Incorrect memories should be deprecated without deleting the audit trail.

7.7 | What the Current Agentic Evidence Actually Supports

Xia et al. (2026) provide an audit-oriented review of agentic trading. Their screened evidence set distinguishes systems that merely predict from those that emit tradable actions and evaluate them in a closed loop. In the primary subset, reporting of chronological splits, explicit transaction costs, universe handling, execution semantics, and reproducible artefacts is sparse. No study in their audit reaches the strongest reproducibility category. This is a substantially more useful conclusion than comparing headline Sharpe ratios across incompatible simulations.

The evidence does not show that multi-agent trading cannot work. It shows that the literature has not yet produced a sufficiently common evaluation protocol to support strong comparative claims. **Architectural novelty is ahead of empirical discipline.**

Table 7 therefore uses an evidence ladder. It separates mature factor findings, established machine-learning applications, promising LLM functions, and experimental agentic systems.

8 | Can Alpha Survive the AI Era?

8.1 | The Cumulative Character of Innovation

The history developed in this survey is cumulative rather than substitutive. Portfolio theory remains relevant because every signal becomes a portfolio. Factors remain relevant because modern models repeatedly rediscover momentum, valuation, liquidity, volatility, and profitability. Alternative data remains relevant because flexible models cannot recover information that was never observed. Execution remains relevant because a prediction earns nothing until a trade is completed. Governance becomes more important as the system becomes harder to understand and easier to scale.

Each generation has exposed an omitted constraint. Mean-variance analysis underestimated input uncertainty. Factor research often underestimated implementation. Alternative-data research can underestimate historical coverage and legality. Machine learning can underestimate selection and cost. LLM systems can underestimate contamination and epistemic reliability. Multi-agent systems can underestimate correlated error and accountability.

The evolution is therefore not a movement away from economics toward computation. The strongest recent work combines flexible estimation with stronger economic and institutional restrictions.

TABLE 7 | Evidence ladder for emerging quantitative methods.

Method or claim	Evidentiary position	What appears useful	What remains unresolved	Appropriate institutional posture
Value, momentum, profitability and low-risk factors	Mature but conditional	Broad economic building blocks and risk decomposition	Timing, valuation and prolonged drawdowns	Use with diversification, cost and exposure controls
Nonlinear ML on structured characteristics	Strong evidence in major studies	Interactions, regularization, ranking and conditional combinations	Transfer across markets, regimes and capital scales	Deploy after strict point-in-time and implementation validation
Latent and deep asset-pricing models	Moderate and growing	Flexible factor extraction under economic structure	Stability, interpretation and mandate-specific superiority	Use where architecture reflects the pricing problem
Portfolio learning with explicit costs	Promising and economically aligned	Connects forecasting to net portfolio utility	Live capacity feedback and cost-model instability	High research priority with conservative production limits
Alternative-data prediction	Dataset specific	Timelier and more granular information	Coverage drift, legal use and post-vendor persistence	Compete on transformation and lineage
LLM extraction, retrieval and embeddings	Promising	Contextual document processing and research productivity	Model-version drift and incremental net alpha	Restrict to bounded, source-grounded tasks
LLM-based return forecasts	Early but non-trivial	Evidence that contextual interpretation can add information	Training contamination, crowding and live implementation	Replicate with frozen models, corpora and timestamps
Autonomous LLM trading	Weak	Experimental platform for decision pipelines	Net returns, capacity, execution and accountability	Do not infer production readiness from backtests
Multi-agent investment systems	Very early	Functional decomposition, critique and memory	Independent diversity, protocol comparability and reproducibility	Treat as governed research infrastructure
Human-AI research organisations	Conceptually compelling	Greater throughput, documentation and systematic challenge	Whether productivity gains become persistent alpha	Measure marginal decision value and preserve human accountability

8.2 | Why Simple Benchmarks Become More Important

Artificial intelligence increases the importance of simple baselines. A complex model can produce many dimensions of apparent improvement, and a weak comparator makes those improvements easy to claim.

The relevant test is

$$\Delta U = U(\text{AI-enhanced feasible system}) - U(\text{best simple feasible system}).$$

The simple system may be $1/N$, a regularised linear model, an equal-weighted factor ensemble, or a slower portfolio with lower turnover. The comparison should charge for data, infrastructure, latency, monitoring, and human review as well as trading costs.

A simple model also serves as a fallback. When a vendor fails, a model drifts, or a generative system becomes unavailable,

the institution needs a safe portfolio state. Robustness includes graceful degradation, not only strong average performance.

8.3 | Where Persistent Alpha May Remain

Alpha can survive because **markets are not homogeneous prediction competitions**. Investors have different liabilities, horizons, regulations, benchmarks, tax positions, financing arrangements, and tolerance for drawdown. A trade unattractive to a highly levered market-neutral fund may be valuable to a pension fund. A signal too small for a large asset manager may support a specialist. A slow risk premium may be unsuitable for short-horizon capital and attractive to patient capital.

Proprietary transformation remains a plausible source of advantage. Raw data increasingly commoditize, but historical corrections, entity graphs, labels, and mappings to economic exposures can remain private.

Execution can remain scarce after prediction becomes common. Funds differ in order routing, crossing, market-impact estimation, inventory management, broker relationships, and the willingness to wait. A known signal does not imply a common realized price.

Economic structure can improve durability. Accounting identities, network propagation, institutional constraints, and causal mechanisms provide a basis for judging whether a relation should survive a new regime. Pattern-only models may forecast better in a stable sample and fail more abruptly when the environment changes.

Organizational learning may be the most durable advantage. A fund that records failed experiments, detects decay early, separates evidence from narrative, and reallocates research effort can repeatedly regenerate alpha even when no individual signal is permanent. Such organizational learning also improves resilience to both recurrent market cycles and less predictable regime shifts.

8.4 | Why Generic AI Alpha Should Decay Quickly

A public foundation model applied to public data through a common prompt is easy to imitate. Model providers can change the representation for all users simultaneously. If many investors use similar models and sources, their forecasts and trades become correlated.

AI therefore has opposing effects: *higher discovery rate and higher commoditisation rate*. The gross number of candidate signals may rise while the average life of a public signal falls. The equilibrium benefit accrues to institutions that connect AI to unique information, cost-aware decisions, and fast feedback—not to those that merely call the newest model.

8.5 | A Research Agenda

The literature needs direct estimates of alpha half-life. Pre- and post-publication averages are informative but coarse. Future work should estimate decay by signal type, market, investor scale, data source, and diffusion channel.

Capacity should become a standard reported outcome. A single cost-adjusted return is insufficient. Researchers should provide net performance across capital, participation rates, borrow conditions, and liquidation horizons.

Alternative-data and LLM studies require point-in-time benchmarks in which document versions, model versions, retrieval corpora, prompts, and timestamps are frozen. Without such infrastructure, later models can appear to predict histories they have already seen.

Machine-learning papers should report the entire decision path from feature to trade. Forecast metrics, turnover, cost assumptions, short feasibility, concentration, factor exposures, and live processing time should be treated as a connected system.

Agentic research needs common protocols before it needs more architectures. Chronological splits, delisting-inclusive universes, explicit transaction costs, order timing, model versions, ablations, and reproducible logs are more valuable at this stage than another collection of specialized agent roles.

Research productivity deserves separate measurement from strategy performance. ***AI may be valuable because it lowers the cost of rejecting bad ideas, not because every AI-generated idea earns a positive return.*** The number of errors prevented, experiments reproduced, and days removed from implementation may be economically meaningful outputs.

9 | Conclusion

Half a century of quantitative equity investing does not culminate in an all-purpose forecasting machine. It culminates in a more demanding definition of what an investment strategy is.

A return predictor becomes alpha only after surviving data reconstruction, model selection, portfolio construction, cost, capacity, latency, regime change, and governance. The new technologies do not remove those stages. They make some stages faster and others more dangerous.

Alpha decay connects the historical generations. Profitable ideas attract attention and capital. Capital changes prices, trading costs, and investor behaviour. Technology accelerates discovery and imitation. The fixed strategy becomes less durable, while the capacity to learn becomes more valuable.

The sustainable advantage of an AI-era quantitative institution can be represented as

$$\text{durable investment value} = \text{information} \times \text{economic structure} \times \text{validation} \times \text{portfolio construction} \times \text{execution} \times \text{governance}$$

The multiplicative form is deliberate. A severe weakness in any component can eliminate the value created by the others.

AI will not end alpha. It will make generic alpha easier to discover, easier to imitate, and harder to retain. The remaining reward will accrue to institutions that possess differentiated data, disciplined skepticism, patient capital, superior execution, and a research organization capable of learning faster than the market adapts.

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Conflicts of Interest

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Generative artificial intelligence tools were used solely to refine the language and improve the clarity of selected sentences, and to produce digital versions of the figures from detailed structures and layouts designed by the author. The conceptual framework, arguments, literature selection, interpretation, figure content, and final verification were undertaken by the author. All AI-assisted text and visual outputs were critically reviewed, revised, and approved by the author, who takes full responsibility for the content of the manuscript.

Data Availability Statement

No new data were collected or generated for this review article. All sources discussed in the manuscript are publicly available through the cited publications.

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